

Original zeta, adelic correspondences, and the full Hurwitz boundary

Programme derivations from the cited human sources

23 September 2026

This reader applies the original Connes–Consani–Marcolli restriction map and trace domain to the programme’s unchanged Hurwitz family and arbitrary split-support lattice. It retains the original zeta and Hurwitz functions, every Gamma and Haar factor, both endpoint values, every boundary residue, all primary-block nilpotents and all support labels. The analytic trace theorem is a precisely cited source input; no new positivity theorem or RH conclusion is asserted.

The first chapter corrects the preceding theta-origin, completion, and fibre explanations. Its maps have their specified domains: meromorphic modules, embedded holomorphic lattices, the transported product, and the common-label support carrier. Scalar zero preservation does not by itself prove preservation of the programme’s multiplication or Weil pairing. The following chapters construct the actual adelic restriction and full boundary receivers.

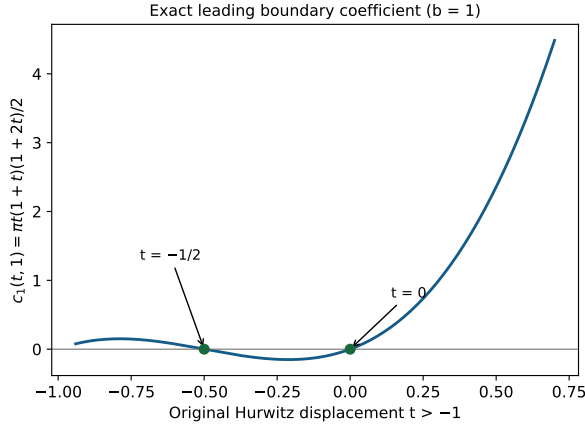
The chapters give the complete new derivations. The source archive also retains the original author archive, the preceding complete programme sources used for comparisons, exact source identities, reproducible figures and supplementary checks. Original-zeta Mellin endpoints and exceptional local lattices remain explicit throughout.

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The exact Hurwitz boundary and its retained support



Every boundary coefficient is retained:

$$c_m(t, b) = \frac{(-1)^{m+1} \pi^m B_{2m+1}(1+t)}{(m-1)! b^{2m}}$$

These are exactly the Mellin residues at $s = -2m$.

$$\partial \Psi_{t,b} = (c_m(t, b))_{m \geq 1}$$

$$\ker \partial = \mathcal{B}$$

The kernel statement includes every derivative.

Mixed cancellation retains a nonzero function and supported zero

$$u = \Psi_{a-1,b} + \Psi_{-a,b} \neq 0, \quad 0 < a < 1, \quad \partial u = 0$$

$$G_L(\partial)(u, 1_L) = (0, 1_L) = e \neq \tau = (0, 0_L)$$

The lattice L is nontrivial. The two marked t values classify individual shifts; reflected sums cancel every boundary term.

Exact formulas and analytic remainder bounds: HCB5-14, HCB18-24, HCB30-33. Human source: Connes-Consani-Marcolli, math/0703392v1, SCK and degmodifv. This plot is not a sampled zero plot or an RH conclusion.

Figure 1: The complete boundary is a sequence, not only its first coordinate. The plotted first coefficient is exact at $b = 1$. Reflected mixtures cancel every coefficient while retaining their nonzero function and supported zero.

Exact arithmetic receivers: endpoint jets and the full primary block

$$\mathcal{M}f_b(s) = 2b^s \zeta(s) \left[\frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right]$$

Endpoints $s = 0$ and $s = 1$

$$\mathcal{M}f_b(0) = 1/4, \quad \mathcal{M}f_b(1) = b/4$$

Finite combinations at distinct widths adjust every finite endpoint jet. The actual restriction-image class and its numerical trace stay fixed.

At an original nontrivial zero ρ of multiplicity m

$$\mathcal{O}/h^{\min(m,N)} \mathcal{O}, \quad h = s - \rho, \quad N[v] = [hv]$$

$$[1] \mapsto [h] \mapsto [h^2] \mapsto 0 \quad (m = N = 3)$$

$$(N^\dagger)^3(v, \lambda) = (0, \lambda) = e \cdot (v, \lambda)$$

WJ17-18 prove the general block and all scaling factors. The three-step chain is the explicitly stated $m = N = 3$ example. Trivial-zero and pole contributions require the retained original multiplier and embedded lattices; no divisor is discarded.

Figure 2: Actual restriction-image functions adjust endpoint jets without changing the spectral class. At a nontrivial zero the full primary block survives, including its nilpotent and its supported-zero power. The displayed three-step example has $m = N = 3$.

1 Original zeta and the retained theta comparison

Correction and provenance. The statement that the Rodgers–Tao heat kernel was the programme’s original construction over the entire absolute base was not established. THS1–31 begin with that classical analytic kernel and construct a support-preserving coefficient lift. HTB1–35 establish its exact comparison with the shifted Hurwitz family. Neither construction is a derivation of the scalar kernel from the entire arithmetic geometry. The distinction is measurable: THS22 already proves that all supported zero times act identically on the actual theta orbit. The ambient coefficient injection does not make that orbit injective.

The original arithmetic functions in this calculation remain

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad F_t(s) = \sum_{n=0}^{\infty} (n+1+t)^{-s}, \quad \operatorname{Re} s > 1, \quad t > -1. \quad (\text{ZTC1})$$

Their continuations, endpoints, and exact shift generator are proved in HTB2–5, with the programme source identified there by version, hash, and original theorem labels. In particular $F_0 = \zeta$, $F_1 = \zeta - 1$, and $\partial_t F_t(s) = -sF_t(s+1)$. The latter family comes from the supplied April 2026 shifted-Hurwitz programme draft, not from a proposed identification of two time variables. Brad Rodgers and Terence Tao’s original author source, equations `phidef`, `htdef`, `hoz`, and `sas`, supplies the auxiliary heat kernel used in THS and HTB. These are cited inputs, not new discoveries. The following comparisons are derived explicitly. Their complete HTB, THS, and CFD proofs accompany this chapter.

1.1 A direct theta sum for the original arithmetic family

For real $t > -1$ and $x > 0$, retain the actual shifted frequencies and put

$$K_t(x) = \sum_{n=0}^{\infty} e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC2})$$

This is a scalar analytic representation of the stated arithmetic family. No assertion that it determines the full support lattice is part of this definition. For fixed $a = 1+t > 0$, monotonicity of the summand as a function of n gives

$$K_t(x) \leq e^{-\pi a^2 x} + \int_0^{\infty} e^{-\pi(a+y)^2 x} dy \leq 1 + \frac{1}{2\sqrt{x}}. \quad (\text{ZTC3})$$

For $x \geq 1$ the same bound after retaining the first exponential gives $K_t(x) \leq C_a e^{-\pi a^2 x}$, since $(a+n)^2 - a^2 \geq n^2$ and $C_a = \sum_{n \geq 0} e^{-\pi((a+n)^2 - a^2)} < \infty$. Consequently the following integral converges absolutely for $\operatorname{Re} s > 1$:

$$\int_0^{\infty} x^{s/2-1} K_t(x) dx = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad F_t(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^{\infty} x^{s/2-1} K_t(x) dx. \quad (\text{ZTC4})$$

Indeed, with $\sigma = \operatorname{Re} s > 1$, the sum of the absolute termwise integrals equals $\pi^{-\sigma/2} \Gamma(\sigma/2) \sum_{n \geq 0} (n+a)^{-\sigma}$. Fubini therefore applies. Substitution $y = \pi(n+a)^2 x$ in each term proves the displayed identity with its full Gamma and pi factors. Continuation outside that domain means the meromorphic continuation proved in HTB4, not an assertion that the unsubtracted integral converges there.

There is also an exact inverse recovering the kernel, not only the arithmetic function. For every real $c > 1$,

$$K_t(x) = \frac{1}{4\pi i} \int_{c-i\infty}^{c+i\infty} \pi^{-s/2} \Gamma(s/2) F_t(s) x^{-s/2} ds. \quad (\text{ZTC4a})$$

Here the integral is absolutely convergent. To prove the inversion with its constant, set $\sigma = c/2 > 0$ and $g(v) = \exp(\sigma v - e^v)$. This function and all its derivatives are integrable on $\mathbb{R}$: at $-\infty$ they are bounded by a constant times $e^{\sigma v}$, and at $+\infty$ by a polynomial in e^v times $g(v)$. The substitution $u = e^v$ identifies its Fourier transform with $\Gamma(\sigma - iy)$. Repeated integration by parts therefore bounds that Gamma factor by $C_N(1 + |y|)^{-N}$ for every fixed N . Fourier inversion gives $e^{-u} = (2\pi i)^{-1} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(w) u^{-w} dw$. One may verify this inversion directly by first multiplying the Fourier integrand by $e^{-\epsilon y^2}$: the resulting expression is the convolution of g with the Gaussian approximate identity, which converges to g ; the stated integrable transform bound permits dominated convergence as $\epsilon \downarrow 0$. Apply the identity to $u = \pi(n + 1 + t)^2 x$, then put $s = 2w$. The resulting factor is $1/(4\pi i)$. Finally sum in n . Interchange is justified by $\sum (n + 1 + t)^{-c} < \infty$ and the integrable Gamma bound. This proves ZTC4a. ZTC4 and ZTC4a are mutual recovery formulas for the specified original arithmetic family and its scalar theta sums; they do not recover a support lattice from a scalar function.

Differentiation in t is locally uniform for t in compact subsets of $(-1, \infty)$ and x in compact subsets of $(0, \infty)$, by Gaussian majorants for a polynomial times e^{-cn^2} . It gives the exact formula

$$\partial_t K_t(x) = -2\pi x \sum_{n=0}^{\infty} (n + 1 + t) e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC5})$$

For $\text{Re } s > 0$, integration of its absolute value is bounded by a constant times $\sum (n + a)^{-\sigma-1}$, locally uniformly in t . More precisely its Mellin transform is

$$\int_0^{\infty} x^{s/2-1} \partial_t K_t(x) dx = -2\pi^{-s/2} \Gamma(s/2 + 1) F_t(s + 1) = -s\pi^{-s/2} \Gamma(s/2) F_t(s + 1). \quad (\text{ZTC6})$$

Thus the original arithmetic shift generator is represented directly, without substituting the heat generator for it. The stated equality extends meromorphically by HTB4–5.

The separated sheet and its entire missing summand are visible before any transform:

$$K_1(x) = K_0(x) - e^{-\pi x}, \quad D_t(s) = \zeta(s) - F_t(s), \quad D_1(s) = 1, \quad (\text{ZTC7})$$

$$\pi^{-s/2} \Gamma(s/2) D_t(s) = \int_0^{\infty} x^{s/2-1} (K_0(x) - K_t(x)) dx \quad (\text{Re } s > 1). \quad (\text{ZTC8})$$

ZTC7 follows by deleting the $n = 1$ summand of K_0 ; ZTC8 follows by subtracting two absolutely convergent integrals. At $t = 1$ the integral is exactly $\pi^{-s/2} \Gamma(s/2)$, so its inverse returns the actual constant 1, including its amplitude and support when lifted.

1.2 Original zeta with both Mellin endpoints retained

At $t = 0$, HTB6 proves

$$K_0(x) = x^{-1/2} K_0(1/x) + \frac{1}{2} x^{-1/2} - \frac{1}{2}. \quad (\text{ZTC9})$$

Split ZTC4 at $x = 1$, apply this identity only to the integral over $(0, 1)$, and substitute $y = 1/x$ there. Initially $\text{Re } s > 1$, and direct integration gives

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \left\{ \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} K_0(x) \left(x^{s/2-1} + x^{-(s+1)/2} \right) dx \right\}. \quad (\text{ZTC10})$$

The two separate rational terms are the transforms of the two separate terms $x^{-1/2}/2$ and $-1/2$. The integral on $[1, \infty)$ is entire in s , since exponential decay bounds every logarithmic derivative on compact subsets of the s -plane. This proves the meromorphic continuation of the identity and keeps both endpoint contributions. Neither has been declared zero or absorbed into an unrecorded constant.

For the original cosine heat kernel, retain

$$\begin{aligned} A(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), & s(z) &= \frac{1}{2} + \frac{iz}{2}, & z(s) &= -2i(s-1/2), \\ H_0(z) &= \frac{s(z)(s(z)-1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2) \zeta(s(z)), \\ \zeta(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_0(z(s)). \end{aligned} \tag{ZTC11}$$

The proof is HTB6–10: the exact Mellin substitution and two integrations by parts yield the displayed multiplier, with all endpoint limits justified in the original convergence half-plane. The last equality is an identity of meromorphic functions. It is not a licence to replace the original zeta calculation by an entire function calculation while dropping its multiplier.

1.3 The boundary is part of the receiving object

Let $\mathcal{M}_s$ and $\mathcal{M}_z$ be the complex vector spaces of meromorphic functions on the respective complex planes. Retain the linear isomorphism, already proved in HTB11,

$$Uf(z) = \frac{A(s(z))}{8} f(s(z)), \quad U^{-1}G(s) = \frac{8G(z(s))}{A(s)}. \tag{ZTC12}$$

Define the joint map on the entire two-component space by

$$\mathcal{V} : \mathcal{M}_s \oplus \mathcal{M}_s \longrightarrow \mathcal{M}_z \oplus \mathcal{M}_z, \quad (F, D) \longmapsto (U(F+D), UD). \tag{ZTC13}$$

It has the exact inverse

$$\mathcal{V}^{-1}(H, J) = \left(\frac{8(H-J) \circ z}{A}, \frac{8J \circ z}{A} \right). \tag{ZTC14}$$

Substitution proves both compositions equal the identity; all entries are meromorphic functions, so no undefined product of separate singular point values is involved. On the actual arithmetic family,

$$\begin{aligned} J_t &= UD_t, & \mathcal{V}(F_t, D_t) &= (H_0, J_t), \\ F_t(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} (H_0(z(s)) - J_t(z(s))), \\ J_1(z) &= \frac{A(s(z))}{8}, & J_t(i) &= -\frac{t}{8}. \end{aligned} \tag{ZTC15}$$

For the last equality use HTB5: $D_t(0) = \zeta(0) - F_t(0) = t$, $A(0) = -1$, and $z(0) = i$. In particular the retained pair determines the real parameter t by $t = -8J_t(i)$. All t give the same first component H_0 . Discarding J_t therefore identifies different actual arithmetic inputs. This is an explicit failure of injectivity on the actual programme family, not merely an example in an unrelated vector space. On the unrestricted space the summation map $(G, J) \mapsto G + J$ has exactly the kernel $\{(B, -B) : B \in \mathcal{M}_z\}$.

For the complete support carrier, use the common label on the direct sum:

$$G_L(\mathcal{M}_s \oplus \mathcal{M}_s) \xrightarrow{\mathcal{V}^\dagger} G_L(\mathcal{M}_z \oplus \mathcal{M}_z), \quad ((F, D), \lambda) \longmapsto (\mathcal{V}(F, D), \lambda). \tag{ZTC16}$$

Here $G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\})$ with its original joins and meets. Additivity, compatibility with the $G_L(\mathbb{C})$ action, carrier membership, and the inverse follow componentwise from linearity and ZTC14, as explicitly proved in CFD11–12. This is a bijection for every bounded distributive L . It does not silently replace the common label by independent labels on two product factors. In particular $e = (0, 1_L)$ and $\tau = (0, 0_L)$ remain distinct when L is nontrivial.

The exact multiplication comparison. The preceding maps were declared as linear and semimodule maps. Their injectivity does not identify the original multiplication with ordinary multiplication of their images. This can be checked exactly, without discarding the completion multiplier. Set

$$B(z) = \frac{A(s(z))}{8} = \frac{s(z)(s(z) - 1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2). \quad (\text{ZTC16a})$$

For meromorphic f, g on the s -plane the two products are

$$\begin{aligned} U(fg)(z) &= B(z)f(s(z))g(s(z)), \\ (Uf)(z)(Ug)(z) &= B(z)^2 f(s(z))g(s(z)). \end{aligned} \quad (\text{ZTC16b})$$

In particular $U1 = B$, not the ordinary constant unit. The factor B is nonconstant: it has a zero at $z(1) = -i$ and poles at $z(-2m)$, $m \geq 1$. Thus ordinary multiplication is not preserved. The exact transported product and its unit are

$$G \odot H = \frac{8G(z)H(z)}{A(s(z))}, \quad 1_{\odot}(z) = \frac{A(s(z))}{8}. \quad (\text{ZTC16c})$$

This formula is an identity of meromorphic functions, including their germs at the zeros and poles of the multiplier. It is not pointwise division of undefined singular values. Substitution gives $(Uf) \odot (Ug) = U(fg)$. Directly, $(G \odot H) \odot K = GHK/B^2 = G \odot (H \odot K)$, $B \odot G = G$, and distributivity follows from distributivity of meromorphic multiplication. Together with ZTC12 this proves a unital algebra isomorphism with the stated product.

For each local holomorphic ring $\mathcal{O}_{s_0}$ the affine coordinate map identifies it with $\mathcal{O}_{z(s_0)}$. Its exact image under U is the embedded fractional lattice $B\mathcal{O}_{z(s_0)}$. This lattice is closed under $\odot$, since $(Bu) \odot (Bv) = Buv$, and has unit B . Keeping instead the unmodified target $\mathcal{O}_{z(s_0)}$ can fail closure: at $s_0 = 1$, B has a simple zero, so $1 \odot 1 = 1/B$ is not holomorphic. This image-lattice statement differs from the preimage lattice $A^{-1}\mathcal{O}$ in ZTC17: the former transports an original holomorphic ring, whereas the latter specifies which original meromorphic sections have a holomorphic completed image.

The full support carrier admits the same exact comparison. Retain its original addition, and define the target product by

$$(G, \lambda) \odot_L (H, \mu) = \left(\frac{8GH}{A \circ s}, \lambda \wedge \mu \right), \quad 1_{\odot_L} = (B, 1_L). \quad (\text{ZTC16d})$$

If the amplitude of a product is nonzero, both input amplitudes are nonzero, hence both labels are 1_L ; this proves carrier closure. If either amplitude is zero its output amplitude is zero and the displayed meet retains its original label. Associativity and distributivity follow from ZTC16c and the meet/join laws of the bounded distributive lattice. The map $(f, \lambda) \mapsto (Uf, \lambda)$ and its inverse therefore give a unital semiring isomorphism from the original coefficient semiring to this specified target, preserving e and τ . An ideal P is sent to its literal image $U^\uparrow(P)$; closure under addition and multiplication follows by applying the isomorphism. Its inverse proves the converse. Moreover a product belongs to that image precisely when the corresponding original product belongs to P , proving that prime ideals correspond. This proves the exact spectrum comparison for these specified coefficient semirings. It is not a derivation of the entire arithmetic base from the scalar function H_0 .

For ZTC13–16, no multiplication on pairs was specified: the proved assertion there remains a common-label semimodule isomorphism. Neither ordinary multiplication nor a Weil pairing is silently inferred from that assertion.

1.4 Local zeros, poles, and the original trace

The earlier conversational assertion that the singular module map had not been established was also too broad: HTB13 already proves $A^{-1}\mathcal{O} \xrightarrow{U} \mathcal{O}$ with the affine coordinate change. CFD1–10

now make the comparison with the *original* regular lattice and every finite jet explicit. At a finite point let $h = s - s_0$ and $A = h^k a(h)$, $a(0) \neq 0$. Retain $I = A^{-1}\mathcal{O} = h^{-k}\mathcal{O}$ as an embedded submodule of the meromorphic germs, and $J = I \cap \mathcal{O}$. For nonzero $f \in I$ the original order is

$$\text{ord}_{s_0} f = \dim_{\mathbb{C}}(I/\mathcal{O}f) - \dim_{\mathbb{C}}(I/J) + \dim_{\mathbb{C}}(\mathcal{O}/J). \quad (\text{ZTC17})$$

This is proved by the power-of- h bases in CFD6–7, without identifying the reference lattices. At a trivial zero $s_0 = -2m$, $I = h\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $\mathcal{O}/I$ has dimension 1. At $s_0 = 1$, $I = h^{-1}\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $I/\mathcal{O}$ has dimension 1. Their signs in ZTC17 give, respectively, +1 and –1. At $s_0 = 0$, $A(0) = -1$ is a unit. CFD4 provides the invertible triangular matrices recovering all Laurent coefficients with the exact factors 8 and $i/2$. Keeping only the zero algebra of the completed section would give zero in both cancelled cases; keeping these comparison modules restores the actual divisor.

The earlier conversational explanation of the fibres requires a further correction. The map

$$\mathcal{O}/h\mathcal{O} \longrightarrow h\mathcal{O}/h^2\mathcal{O}, \quad [u] \longmapsto [hu] \quad (\text{ZTC17a})$$

is an isomorphism of $\mathcal{O}$ -modules. It is well-defined because $h(h\mathcal{O}) = h^2\mathcal{O}$, surjective by the definition of $h\mathcal{O}$, and injective because $hu \in h^2\mathcal{O}$ implies $u \in h\mathcal{O}$ in the local integral domain. Its inverse is $[hu] \mapsto [u]$. Consequently the two abstract fibres alone do not account for the loss. What must be retained is their specified embedding and the original reference lattice: multiplication by $A = h^{-1}a(h)$ identifies $I = h\mathcal{O}$ with $\mathcal{O}$, but the length-one quotient $\mathcal{O}/I$ records the cancelled original zero. Replacing the entire marked comparison by the completed section's zero quotient discards precisely that datum. Likewise multiplication by nonzero meromorphic A is an automorphism of the meromorphic sheaf as a module. It is not a ring automorphism for ordinary multiplication, as ZTC16b proves. It is not an automorphism of the fixed holomorphic lattice at its zeros and poles. These category statements do not contradict one another.

Logarithmic differentiation gives the original trace identity

$$\frac{\zeta'(s)}{\zeta(s)} = -2i \frac{H'_0(z(s))}{H_0(z(s))} - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \log \pi - \frac{1}{2} \psi(s/2). \quad (\text{ZTC18})$$

It first holds away from zeros and poles and then meromorphically; $\psi = \Gamma'/\Gamma$. For a positively oriented finite contour γ avoiding all zeros and poles of the functions and all singularities of every displayed integrand, and a holomorphic test g on and inside it, multiplication by g and integration gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} g \frac{\zeta'}{\zeta} ds &= \frac{1}{2\pi i} \int_{\gamma} g(s) (-2i) \frac{H'_0(z(s))}{H_0(z(s))} ds \\ &\quad - \frac{1}{2\pi i} \int_{\gamma} g(s) \left(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) \right) ds. \end{aligned} \quad (\text{ZTC19})$$

At every zero or pole, write a meromorphic function as $(s - s_0)^r$ times a holomorphic unit. Its logarithmic derivative then has residue r . The residue theorem proves that ZTC19 counts the original zeros with multiplicity and the original poles with negative multiplicity, including every trivial zero inside the contour. No infinite-contour limit or decay assumption is hidden here. Extending this finite identity to a particular infinite trace requires the actual convergence estimates of that trace.

1.5 Two different trajectories through the same original zeta

The original- s function associated with the classical heat path is

$$f_h^{\text{heat}}(s) = \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_h(z(s)), \quad f_0^{\text{heat}} = \zeta. \quad (\text{ZTC20})$$

CFD18–21 give its full receiving operator and prove

$$\partial_h f_h^{\text{heat}}(-2)\Big|_{h=0} = 0, \quad \partial_t F_t(-2)|_{t=0} = -\frac{1}{6}. \quad (\text{ZTC21})$$

For completeness, the heat equation and chain rule give $\partial_h f = (Af)''/(4A)$. At $h = 0$, $A\zeta$ is holomorphic at -2 whereas $1/A$ has a simple zero there, giving the first value. HTB5 gives the second as $2\zeta(-1) = -1/6$. The discrepancy holds on the actual initial arithmetic function. CFD21 further proves that all the zeros at $-2m$ remain simple along the real heat path, while its residue at 1 varies strictly; the actual Hurwitz path has constant residue 1 and values $-B_{2m+1}(1+t)/(2m+1)$ at those points.

Thus the valid original- ζ receiver consists of the original functions, their exact transform and inverse, the retained boundary component, the embedded local comparison modules, and all support labels. The scalar theta orbit by itself is a smaller observable. These proofs neither exhibit an off-line zero of the original ζ nor rule one out. They establish precisely which input and which deformation a subsequent estimate is estimating.

2 The original trivial-correspondence family with every Haar constant

Human source and reading scope. Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author file `Yuri70v2.tex`, subsection *Fubini's theorem and the trivial correspondences*, especially Lemma `degmodifyV`, supply the polynomial Gaussian used below. The original file has SHA-256 `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`. The complete subsection, source lines 2313–2424, was read, together with the definitions `fsharpinv`, `degcorrZf`, and `codegcorrZf`. This calculation fixes the measures and sums over all nonzero rational numbers; it replaces the source's unspecified constant by a proved exact value. It does not replace the original zeta function with a completed function, and makes no RH inference.

2.1 Original adelic test, rational restrictions, and additive moments

Use additive Haar measure dx at infinity and $\text{vol}(\mathbb{Z}_p) = 1$ at every finite prime. On $C_{\mathbb{Q}} = \widehat{\mathbb{Z}}^{\times} \times \mathbb{R}_{>0}$ use Haar measure $du dy/y$, with $\text{vol}(\widehat{\mathbb{Z}}^{\times}) = 1$. Retain

$$\eta(x) = \pi x^2 \left(\pi x^2 - \frac{3}{2} \right) e^{-\pi x^2} = \left(\pi^2 x^4 - \frac{3\pi}{2} x^2 \right) e^{-\pi x^2}, \quad k \in \bigoplus_p \mathbb{Z}, \quad N(k) = \prod_p p^{k_p}, \quad a > 0, \quad (\text{ETC1})$$

and the actual tensor

$$\xi_{k,a}(x) = \eta(x_{\infty}/a) \prod_p 1_{p^{k_p} \mathbb{Z}_p}(x_p), \quad b = \frac{a}{N(k)} > 0. \quad (\text{ETC2})$$

The symbol b records this specified ratio; it does not change either of the original parameters or the local factors. The finite product of nonstandard factors defines a Bruhat–Schwartz function on the original adèles.

The Gaussian integral and its two differentiated forms are

$$\int_{\mathbb{R}} e^{-\pi x^2} dx = 1, \quad \int_{\mathbb{R}} x^2 e^{-\pi x^2} dx = \frac{1}{2\pi}, \quad \int_{\mathbb{R}} x^4 e^{-\pi x^2} dx = \frac{3}{4\pi^2}. \quad (\text{ETC3})$$

For example, square the first integral and use polar coordinates to obtain one; differentiating $\int e^{-\pi t x^2} dx = t^{-1/2}$ once and twice at $t = 1$ gives the other two. Polynomial Gaussian bounds justify both differentiations. Consequently

$$\eta(0) = 0, \quad \int_{\mathbb{R}} \eta = \frac{3}{4} - \frac{3}{4} = 0, \quad \xi_{k,a}(0) = 0, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi_{k,a} = a \prod_p p^{-k_p} \left(\frac{3}{4} - \frac{3}{4} \right) = 0. \quad (\text{ETC4})$$

Thus the two additive zero-moment conditions hold on the original test, with both summands and every volume retained.

Define the reduction by the original full rational sum:

$$f_{k,a}(u, y) = (\mathcal{E}\xi_{k,a})(u, y) = \sum_{q \in \mathbb{Q}^{\times}} \xi_{k,a}(qy, (qu_p)_p). \quad (\text{ETC5})$$

Since u_p is a unit, its finite characteristic factors permit exactly $v_p(q) \geq k_p$ at every prime. This says that $q/N(k)$ is a rational number integral at every finite prime, hence an integer: a reduced denominator greater than one would give a negative valuation at one of its prime divisors. Conversely every nonzero integer satisfies these conditions. The map $n \mapsto N(k)n$ is therefore a bijection onto the permitted rational numbers. Since η is even,

$$f_{k,a}(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \eta \left(\frac{N(k)ny}{a} \right) = 2 \sum_{n=1}^{\infty} \eta(ny/b) =: f_b(y). \quad (\text{ETC6})$$

The factor two is from the two rational signs. This is an exact map of the specified family; it shows explicitly that this scalar reduction depends on k, a only through $a/N(k)$. It does not recover the individual finite-place factors from that ratio.

2.2 Fourier calculation, Poisson identity, and convergence

Use the real Fourier convention $\widehat{h}(v) = \int h(x)e^{-2\pi i x v} dx$ and compatible conductor $\mathbb{Z}_p$ characters at finite places. Let $g(x) = e^{-\pi x^2}$. The Fourier integral has derivative $-2\pi v \widehat{g}(v)$: differentiate under the integral and integrate $g' = -2\pi x g$ by parts. Its value at zero is one by ETC3, so $\widehat{g}(v) = g(v)$. Differentiating this identity gives the complete two polynomial transforms

$$\begin{aligned}\widehat{x^2 g}(v) &= \left(\frac{1}{2\pi} - v^2\right) g(v), \\ \widehat{x^4 g}(v) &= \left(v^4 - \frac{3}{\pi} v^2 + \frac{3}{4\pi^2}\right) g(v), \\ \widehat{\eta}(v) &= \left[\pi^2 \left(v^4 - \frac{3}{\pi} v^2 + \frac{3}{4\pi^2}\right) - \frac{3\pi}{2} \left(\frac{1}{2\pi} - v^2\right)\right] g(v) = \eta(v).\end{aligned}\tag{ETC7}$$

These differentiations and integrations by parts have vanishing boundary terms because every derivative is a polynomial times a Gaussian. At a finite prime, character orthogonality gives $\widehat{1_{p^{k_p}\mathbb{Z}_p}} = p^{-k_p} 1_{p^{-k_p}\mathbb{Z}_p}$: the transform equals the original ball volume on its annihilator and is zero outside it. The real dilation contributes a . Hence

$$\widehat{\xi}_{k,a} = a \prod_p p^{-k_p} \xi_{-k,1/a} = \frac{a}{N(k)} \xi_{-k,1/a}.\tag{ETC8}$$

Applying Fourier twice yields the original even test: the second coefficient is $(1/a)/N(-k) = N(k)/a$, so the two displayed coefficients multiply to one.

For the required Poisson identity no rearrangement of a divergent integral is used. For $r > 0$, periodize $\eta(r(x+n))$ over $n \in \mathbb{Z}$. Every differentiated series converges uniformly on $x \in [0, 1]$ by Schwartz bounds. Its m th Fourier coefficient is $r^{-1} \widehat{\eta}(m/r)$, by the absolutely convergent substitution over all intervals. These coefficients decrease faster than any power, so their Fourier series converges absolutely to the periodization. Evaluating at zero and using ETC7 gives

$$\sum_{n \in \mathbb{Z}} \eta(nr) = \frac{1}{r} \sum_{m \in \mathbb{Z}} \eta(m/r), \quad f_b(y) = \frac{b}{y} f_{1/b}(1/y).\tag{ETC9}$$

The zero summand is zero on both sides because $\eta(0) = 0$. The same argument with g proves its Gaussian Poisson formula.

For $y \geq 1$ the series in ETC6 and each of its derivatives are bounded by a polynomial in y times $e^{-\pi y^2/b^2}$. Indeed factor this last exponential from each term; the remaining sum is bounded by a convergent series of a polynomial in n times $e^{-\pi(n^2-1)/b^2}$. ETC9 therefore bounds f_b near zero by a constant times $y^{-5} e^{-\pi b^2/y^2}$; its differentiated versions have the same exponential with possibly different finite powers of y^{-1} . Thus this actual reduced test and all its multiplicative derivatives decrease faster than every power at both ends. In particular every Mellin integral below converges absolutely for every s , locally uniformly with all s derivatives.

2.3 Original-zeta Mellin formula with both Gaussian summands

With the specified class-group measure put

$$M_b(s) = \int_{\widehat{\mathbb{Z}}^\times} du \int_0^\infty f_b(y) y^s \frac{dy}{y} = \int_0^\infty f_b(y) y^{s-1} dy.\tag{ETC10}$$

The preceding bounds prove that M_b is entire. Initially for $\operatorname{Re} s > 1$, absolute Fubini is valid on ETC6, since

$$2 \sum_{n \geq 1} \int_0^\infty |\eta(ny/b)| y^{\operatorname{Re} s - 1} dy = 2b^{\operatorname{Re} s} \sum_{n \geq 1} n^{-\operatorname{Re} s} \int_0^\infty |\eta(x)| x^{\operatorname{Re} s - 1} dx < \infty.$$

The local integral converges already for $\operatorname{Re} s > -2$, as $\eta(x) = O(x^2)$ at zero. Substituting $v = \pi x^2$ into each of the two original polynomial-Gaussian terms proves

$$\begin{aligned} \int_0^\infty \eta(x) x^{s-1} dx &= \frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right), \\ M_b(s) &= 2 \left(\frac{a}{N(k)}\right)^s \zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right]. \end{aligned} \quad (\text{ETC11})$$

The working expression keeps the original $\zeta(s)$, both Gamma summands, the two rational signs, and all finite and real factors. It continues meromorphically by the original zeta continuation ZTC10; equality with the entire integral ETC10 proves that its singularities are removable as a whole, without asserting that its separate summands are entire.

For a precise comparison with the source's unspecified constant, the two-step Gamma recurrence gives the identity

$$2 \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right] = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2). \quad (\text{ETC12})$$

This is an explicit equality of the retained multiplier, not a replacement arithmetic function. For real u it yields the exact source comparison

$$M_b(iu) = -\frac{b^{iu}}{4} u(u+i) \pi^{-iu/2} \Gamma(iu/2) \zeta(iu). \quad (\text{ETC13})$$

At removable points its value is the entire continuation specified by ETC10. The factor $-1/4$ is for the full rational sum in ETC5. Using only the positive integers would instead give $-1/8$ and would not be the reduction defined in ETC5.

2.4 Both endpoint values without an illicit interchange

Retain the complete Gaussian theta sum, including its zero term:

$$\Theta_b(y) = \sum_{n \in \mathbb{Z}} e^{-\pi(ny/b)^2}, \quad f_b(y) = \frac{1}{4} (y^2 \Theta_b''(y) + 2y \Theta_b'(y)) = \frac{1}{4} (y^2 \Theta_b'(y))'. \quad (\text{ETC14})$$

For each summand this follows from $x^2 g''(x) + 2x g'(x) = 4\eta(x)$; the differentiated sums converge locally uniformly. Gaussian Poisson summation as proved after ETC9 gives $\Theta_b(y) = (b/y) \Theta_{1/b}(1/y)$. As $y \rightarrow \infty$, $\Theta_b(y) \rightarrow 1$ and $y \Theta_b'(y), y^2 \Theta_b''(y) \rightarrow 0$. As $y \downarrow 0$, its leading term is b/y , with an exponentially small remainder and differentiated remainders. Consequently

$$\begin{aligned} \lim_{y \downarrow 0} y^2 \Theta_b'(y) &= -b, \\ \lim_{y \downarrow 0} (y \Theta_b'(y) + \Theta_b(y)) &= 0, \\ M_b(1) &= \frac{1}{4} [y^2 \Theta_b'(y)]_0^\infty = \frac{b}{4} = \frac{a}{4N(k)}, \\ M_b(0) &= \frac{1}{4} [y \Theta_b'(y) + \Theta_b(y)]_0^\infty = \frac{1}{4}. \end{aligned} \quad (\text{ETC15})$$

Integrate first on a compact interval $[\epsilon, R]$ and then take the endpoints; ETC9–10 guarantee convergence of the integrals, and the displayed limits justify both passages. Thus the two endpoint

values are calculated from the actual reduced test, without assuming special zeta values and without exchanging a nonabsolutely convergent sum and integral.

The failure of the tempting degree interchange is quantified exactly. Each $n \neq 0$ term has integral zero in dy , by ETC4, but

$$\sum_{n \neq 0} \int_0^\infty |\eta(ny/b)| dy = 2b \left(\int_0^\infty |\eta(x)| dx \right) \sum_{n=1}^\infty \frac{1}{n} = \infty. \quad (\text{ETC16})$$

The coefficient integral is strictly positive because η is continuous and not identically zero. The codegree interchange also lacks absolute convergence: each absolute integral with dy/y equals the same positive finite number $\int_0^\infty |\eta(x)| dx/x$, so its sum diverges. This does not conflict with the convergent whole-test endpoint values ETC15.

2.5 Sharp, conjugation, and two independent endpoint adjustments

Use the source's literal involution

$$f^\#(u, y) = y^{-1} \overline{f(u^{-1}, y^{-1})}. \quad (\text{ETC17})$$

The tests f_b are real and independent of u . Applying ETC9 at $1/y$, and comparing with ETC8, proves

$$f_b^\#(y) = b f_{1/b}(y) = \mathcal{E} \widehat{\xi}_{k,a}(y), \quad M_{f_b^\#}(s) = b M_{1/b}(s) = M_b(1-s). \quad (\text{ETC18})$$

For a general complex linear combination f of these tests, the exact formula instead is

$$M_{f^\#}(s) = \overline{M_f(1-\bar{s})}, \quad M_{f^\#}(1) = \overline{M_f(0)}. \quad (\text{ETC19})$$

Indeed substitute $v = 1/y$ in the absolutely convergent integral of $y^{-1} \overline{f(1/y)} y^{s-1}$; its integrand becomes $\overline{f(v)} v^{-s}$, which is the displayed conjugated Mellin transform. The source writes codegree both as $M_f(0)$ and as the degree of $f^\#$. Those values agree on its real test functions, including every individual f_b here; for complex coefficients the latter is the conjugate of the former. The formulas below specify which convention is being adjusted.

Define the complex-linear endpoint pair as $(d(f), c(f)) = (M_f(1), M_f(0))$. For two positive widths $b_1 \neq b_2$, both in the actual original family, ETC15 gives

$$\begin{pmatrix} d(c_1 f_{b_1} + c_2 f_{b_2}) \\ c(c_1 f_{b_1} + c_2 f_{b_2}) \end{pmatrix} = \frac{1}{4} \begin{pmatrix} b_1 & b_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad \det \left(\frac{1}{4} \begin{pmatrix} b_1 & b_2 \\ 1 & 1 \end{pmatrix} \right) = \frac{b_1 - b_2}{16} \neq 0. \quad (\text{ETC20})$$

Thus prescribed changes $(D, C) \in \mathbb{C}^2$ are attained by the explicit inverse

$$c_1 = \frac{4(D - b_2 C)}{b_1 - b_2}, \quad c_2 = \frac{4(b_1 C - D)}{b_1 - b_2}. \quad (\text{ETC21})$$

Substitution gives $b_1 c_1 + b_2 c_2 = 4D$ and $c_1 + c_2 = 4C$, proving the inverse without a dimension argument. Both widths may use the same arbitrary divisor k by choosing $a_i = N(k) b_i$. The actual antecedent $c_1 \xi_{k,a_1} + c_2 \xi_{k,a_2}$ still has zero value and zero additive integral by ETC4, and its reduction is the displayed sum. For real desired changes the coefficients are real. If instead “codegree” literally means $d(f^\#)$ for complex f , replace C by $\bar{C}$ in ETC21; ETC19 then proves the corresponding real-linear inverse. This records the conjugation rather than erasing it.

Finally the original adelic rational action is

$$(\alpha_r \xi)(x) = \xi(r^{-1}x), \quad \alpha_r \xi_{k,a} = \xi_{k+d(r), |r|_\infty a}, \quad N(k + d(r)) = |r|_\infty N(k), \quad \mathcal{E} \alpha_r \xi_{k,a} = \mathcal{E} \xi_{k,a}. \quad (\text{ETC22})$$

The finite balls shift by their exact valuations; evenness of η removes the real sign and leaves the real factor $|r|_\infty$. The equality of N follows from unique factorization of a nonzero rational

number. The last equality follows either from the retained ratio in ETC6 or by reindexing the full rational sum $q \mapsto q/r$. Real scaling alone changes a to λa , and hence $f_b(y)$ to $f_{\lambda b}(y) = f_b(y/\lambda)$: its Mellin transform acquires exactly λ^s , its degree exactly λ , and its codegree remains unchanged. These are the actual two endpoint coordinates used in ETC20–21, with every prime and real factor retained. The calculation concerns this image family and its exact adjustment maps; it does not assert positivity of a Weil pairing.

There is a two-dimensional adjustment space stable under the literal sharp involution. Choose any $b > 0$ with $b \neq 1$ and put $c = 1/b$. In the basis (f_b, f_c) the conjugate-linear operation is exactly

$$(z_1 f_b + z_2 f_c)^\sharp = \frac{\bar{z}_2}{b} f_b + b \bar{z}_1 f_c, \quad \begin{pmatrix} 0 & 1/b \\ b & 0 \end{pmatrix}^2 = I. \quad (\text{ETC23})$$

Both coefficients come from ETC18, including its width factor. Define in this same space the original linear combinations

$$A_d = \frac{4(f_b - f_c)}{b - c}, \quad A_c = \frac{4(bf_c - cf_b)}{b - c}. \quad (\text{ETC24})$$

Direct substitution into ETC15 proves $(d(A_d), c(A_d)) = (1, 0)$ and $(d(A_c), c(A_c)) = (0, 1)$. The determinant in ETC20 proves that these moments uniquely determine an element of this two-dimensional space. ETC19 and its invariance under sharp therefore give $A_d^\sharp = A_c$ and $A_c^\sharp = A_d$, or equivalently

$$(DA_d + CA_c)^\sharp = \bar{C}A_d + \bar{D}A_c. \quad (\text{ETC25})$$

These are explicit combinations of the two original tests and their actual rational reductions. In particular they supply every prescribed endpoint pair while retaining the involution and the original zero-additive-moment antecedents.

For completeness, the two Gamma contributions in ETC11 have the following separate behavior at the two endpoint arguments. Write $h = s - 1$ and $\zeta(s) = h^{-1} + c_0 + O(h)$, where c_0 denotes its retained Laurent constant. Set $L_b = \log b - \frac{1}{2} \log \pi$. The first Gamma summand including its factors has expansion

$$\begin{aligned} b^s \pi^{-s/2} \Gamma(s/2 + 2) \zeta(s) &= \frac{3b}{4h} + \frac{3b}{4}(c_0 + L_b) + \frac{b}{2\sqrt{\pi}} \Gamma'(5/2) + O(h), \\ -\frac{3}{2} b^s \pi^{-s/2} \Gamma(s/2 + 1) \zeta(s) &= -\frac{3b}{4h} - \frac{3b}{4}(c_0 + L_b) - \frac{3b}{4\sqrt{\pi}} \Gamma'(3/2) + O(h). \end{aligned} \quad (\text{ETC26})$$

The Gamma recurrence differentiated at $3/2$ gives $\Gamma'(5/2) = \frac{3}{2}\Gamma'(3/2) + \Gamma(3/2)$. Adding the two displayed expansions therefore retains and cancels their opposite residues and their equal Laurent and width constants, leaving exactly $b/4$. At $s = 0$ the two summands have values $\zeta(0)$ and $-\frac{3}{2}\zeta(0)$. Their bracket is $-1/2$; ETC15 and ETC11 imply $\zeta(0) = -1/2$, so their separate values are $-1/2$ and $3/4$. Their sum is the independently calculated $1/4$. Thus neither endpoint calculation has discarded the contribution of one original Gaussian term.

3 The original adelic trace quotient with every support label retained

Human sources, objects, and reading scope. The analytic input is Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author file `Yuri70v2.tex`, SHA-256 `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`, canonical programme source `PUBUNIT-5DE2A2D44F8B1470AC701FDA`. The sections actually read for this derivation are source lines 610–710, 1316–1425, 1508–1551, 1680–1830, 1980–2160, 2161–2216, and 2313–2424, together with the title and author lines. The original locators are `SCK`, `rangeTrpi`, `rangeCvdef`, `varthetaH1V`, `varthetafaconvol`, `vanishV`, `tracepairing`, `fsharpinv`, `TraceformulaThm`, and `TraceformulaH1`. The paragraph following `moth1AKCK` specifies closure of the range in the cokernel. The trace-class and explicit-formula theorems are cited analytic inputs. The convolution ideal, its sharp invariance with all endpoint terms, and the full-label quotient and receiving maps are derived below. No positivity assertion or RH conclusion is made.

3.1 The original test algebra and its domains

Retain the source’s unscaled test algebra and operations:

$$C = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}, \quad B = \bigcap_{\beta \in \mathbb{R}} |\cdot|^{\beta} \mathcal{S}(C), \quad (f * g)(x) = \int_C f(h)g(h^{-1}x) d^{\times}h, \quad f^{\sharp}(x) = |x|^{-1} \overline{f(x^{-1})}. \quad (\text{SWQ1})$$

For $\mathbb{Q}$, the actual continuous multiplicative section in the author’s subsection `XXCKsect` is

$$C = \mathbb{R}_{>0} \times K, \quad K = \widehat{\mathbb{Z}}^{\times}, \quad (r, u) \mapsto (r, (u_p)_p), \quad |(r, u)| = r. \quad (\text{SWQ2})$$

Indeed one multiplies an idele by the unique rational number that makes every finite component a unit and its real component positive. The ambiguity would have all finite valuations zero and positive real component, so would be the rational number 1. Multiplication respects these representatives. The multiplicative Haar measure is $dr/r du$, where $\int_K du = 1$. The additive adelic measure is real Lebesgue measure times the measures of volume 1 on each $\mathbb{Z}_p$. These exact measures are retained throughout.

The topology used for closure below is the usual weighted Bruhat test-function topology: at each fixed open subgroup $H \subset K$, functions are H -invariant and have all weighted Schwartz seminorms; one then uses the Bruhat inductive-limit topology over these finite levels. An explicit defining family at a fixed level, with $r = e^x$, is

$$p_{N,m,j}(f) = \sup_{x \in \mathbb{R}, u \in K} e^{N|x|} (1 + |x|)^m |\partial_x^j f(e^x, u)|, \quad N, m, j \in \mathbb{Z}_{\geq 0}. \quad (\text{SWQ3})$$

These seminorms retain all real weights in SWQ1: weights $e^{\beta x}$ are bounded by $e^{N|x|}$ when $N \geq |\beta|$, and conversely the two weights e^{Nx} and e^{-Nx} control $e^{N|x|}$; Leibniz’s formula controls their derivatives. This is a description of the original test domain, not a change of function or scale in its operations.

All convolution integrals in SWQ1 converge absolutely. Using $x = y + (x - y)$ and $1 + |x| \leq (1 + |y|)(1 + |x - y|)$ gives the explicit bound

$$p_{N,m,j}(f * g) \leq 2p_{N+1,m,0}(f) p_{N,m,j}(g). \quad (\text{SWQ4})$$

The constant 2 is $\int_{\mathbb{R}} e^{-|y|} dy$; the compact Haar factor is 1. Differentiation can pass under the integral by the same bound on every derivative. Convolution preserves a common finite level,

and this proves the continuity needed below. Since C is abelian, substitution $h \mapsto xh^{-1}$ proves commutativity. Fubini, justified by the corresponding three-function L^1 bound, proves associativity.

In these coordinates sharp is exactly

$$f^\sharp(e^x, u) = e^{-x} \overline{f(e^{-x}, u^{-1})}, \quad p_{N,m,j}(f^\sharp) \leq \sum_{\ell=0}^j \binom{j}{\ell} p_{N+1,m,\ell}(f). \quad (\text{SWQ5})$$

The bound follows by differentiating e^{-x} and $f(e^{-x}, u^{-1})$; the two derivative signs together have absolute value 1. Direct substitution gives $(f^\sharp)^\sharp = f$. Inversion preserves Haar measure on this abelian group, and the norm is a character. Substitution in the absolutely convergent convolution therefore gives $(f * g)^\sharp = g^\sharp * f^\sharp$. In particular sharp is continuous on the stated test space. The algebra B has no convolution unit among its functions. Indeed let $b(e^x, u) = e^{-x^2}$, which belongs to B and has trivial compact-unit character. Its real Fourier transform is $\sqrt{\pi} e^{-t^2/4}$, nonzero for every real t . A convolution unit $a \in B$ would consequently have $\int_{\mathbb{R} \times K} a(e^x, u) e^{itx} dx du = 1$ for every t . Integration by parts in x , justified by SWQ3, makes this integral tend to zero as $|t| \rightarrow \infty$, a contradiction. No unit distribution is inserted in this chapter.

3.2 The adelic source, with both Poisson endpoints

Use the canonical additive character trivial on $\mathbb{Q}$, with real component $e^{-2\pi ixy}$ and finite conductor $\mathbb{Z}_p$, and its self-dual measures specified after SWQ2. Write

$$\begin{aligned} \widehat{\xi}(y) &= \int_{\mathbb{A}} \xi(x) \alpha(xy) dx, & \varepsilon_0(\xi) &= \xi(0), & \varepsilon_1(\xi) &= \int_{\mathbb{A}} \xi(x) dx, \\ \mathcal{S}(\mathbb{A})_0 &= \ker \varepsilon_0 \cap \ker \varepsilon_1, & (E\xi)(g) &= \sum_{q \in \mathbb{Q}^\times} \xi(qg), & V &= E\mathcal{S}(\mathbb{A})_0. \end{aligned} \quad (\text{SWQ6})$$

For the reference characters e_v in SWQ28, choose exactly these local components of α . This fixes more than their conductors: it specifies the characters themselves, so that the differential idele there is $a_v = 1$ at every place. The sum is invariant under the rational choice of representative. For fixed g , finite-place compact support puts its contributing rational q in $D^{-1}\mathbb{Z}$ for one positive integer D . Real Schwartz decay then makes the sum absolutely convergent, locally uniformly with every real derivative. Unit dilation at finite places preserves the required common denominator and bounds.

For completeness the full Poisson formula used here is

$$\sum_{q \in \mathbb{Q}} \xi(qg) = |g|^{-1} \sum_{q \in \mathbb{Q}} \widehat{\xi}(qg^{-1}). \quad (\text{SWQ7})$$

It follows directly for the original Bruhat space. A finite-place test function is a finite linear combination of characteristic functions of cosets $a + N\widehat{\mathbb{Z}}$, with rational a and positive integer N , after a common denominator is chosen. The rational elements in such a coset are $a + N\mathbb{Z}$. Real periodization of a Schwartz function gives

$$\sum_{n \in \mathbb{Z}} \phi(a + Nn) = \frac{1}{N} \sum_{m \in \mathbb{Z}} e^{2\pi i m a / N} \widehat{\phi}(m/N).$$

To verify this last identity, periodize over $N\mathbb{Z}$, integrate over one period to compute every Fourier coefficient, and evaluate the resulting absolutely convergent Fourier series at a . The finite Fourier transform of the characteristic coset is $N^{-1} \alpha_f(ay) 1_{N^{-1}\widehat{\mathbb{Z}}}(y)$. At rational $y = m/N$, its character is $e^{2\pi i a m / N}$ because the global character is trivial on rational numbers. These exact factors prove adelic Poisson by linearity. Finally replacing the test function by $x \mapsto \xi(gx)$ multiplies its Fourier transform by $|g|^{-1}$ and evaluates it at $g^{-1}y$, proving SWQ7. The same real Fourier inversion and finite character orthogonality give $\widehat{\widehat{\xi}}(x) = \xi(-x)$.

Separating the actual $q = 0$ terms in SWQ7, with neither endpoint discarded, gives

$$E\xi(g) = |g|^{-1}E\widehat{\xi}(g^{-1}) + |g|^{-1}\varepsilon_1(\xi) - \varepsilon_0(\xi). \quad (\text{SWQ8})$$

For $|g| \geq 1$, the earlier common-denominator estimate gives $E\xi(g) = O_M(|g|^{-M})$ for every M , uniformly in the compact unit coordinate and with every logarithmic real derivative. For $\xi \in \mathcal{S}(\mathbb{A})_0$, both displayed endpoints vanish by definition, and the right side applies the same large-norm estimate to $\widehat{\xi}$ at g^{-1} . Indeed $\widehat{\xi}(0) = \varepsilon_1(\xi) = 0$ and $\int \widehat{\xi} = \xi(0) = 0$ by Fourier inversion. The factor $|g|^{-1}$ stays present. Since M is arbitrary, it proves all the small-norm bounds in SWQ3 as well. Thus $V \subset B$ on the source's actual domain.

Apply SWQ8 to $\bar{\xi}$, or equivalently apply SWQ7 to $\bar{\xi}$ and invert g . The exact result is

$$(E\xi)^\sharp(g) = E\widehat{\bar{\xi}}(g) - |g|^{-1}\overline{\varepsilon_0(\xi)} + \overline{\varepsilon_1(\xi)}. \quad (\text{SWQ9})$$

This identity retains the conjugations, the factor $|g|^{-1}$, and both endpoints. Put $J\xi = \widehat{\bar{\xi}}$. Its endpoints are

$$\varepsilon_0(J\xi) = \overline{\varepsilon_1(\xi)}, \quad \varepsilon_1(J\xi) = \overline{\varepsilon_0(\xi)}. \quad (\text{SWQ10})$$

Consequently J preserves the actual space $\mathcal{S}(\mathbb{A})_0$ and SWQ9 gives $V^\sharp \subset V$; applying sharp twice gives equality. This derives sharp invariance from additive Fourier–Poisson, rather than assuming that the original quotient was an involutive quotient.

3.3 The actual convolution ideal and its closure

For $f \in B$, $\xi \in \mathcal{S}(\mathbb{A})_0$, use the actual multiplicative section SWQ2 to define

$$\eta(x) = \int_C f(h)\xi(h^{-1}x) d^\times h. \quad (\text{SWQ11})$$

This integral belongs to the original Bruhat–Schwartz space. Indeed the orbit of the finite-place component of ξ under the compact group K lies in a finite-dimensional space of test functions: its compact support stays in one finite product of bounded balls, and its common local-constancy subgroups are unchanged by units. The finite quotient of this compact support by these subgroups has finitely many points. For the real component, every Schwartz seminorm has the explicit bound

$$\sup_{x_\infty, x_f} |x_\infty^N \partial_{x_\infty}^j \eta(x)| \leq C_{\xi, N, j} \int_C |f(h)| |h|^{N-j} d^\times h < \infty. \quad (\text{SWQ12})$$

The last integral is finite by SWQ3. Differentiation and integration are justified by the same bound, while the fixed finite-place test space makes the resulting function a Bruhat test function.

The two endpoints of this integral are exactly

$$\eta(0) = \varepsilon_0(\xi) \int_C f(h) d^\times h, \quad \int_{\mathbb{A}} \eta(x) dx = \varepsilon_1(\xi) \int_C |h| f(h) d^\times h. \quad (\text{SWQ13})$$

For the second equation the absolute double integral is bounded by $\|\xi\|_{L^1(\mathbb{A})} \int_C |h| |f(h)| d^\times h$, which is finite. The change of variables $x = hy$ therefore gives its exact adelic factor $|h|$, and Fubini is valid for this equation. Thus $\eta \in \mathcal{S}(\mathbb{A})_0$.

There is also a justified interchange with the arithmetic sum in the convolution calculation. For fixed g , real Schwartz decay and the fixed rational denominator give, uniformly in the unit part of h , the estimate

$$\sum_{q \neq 0} |\xi(qh^{-1}g)| \leq C_{\xi, g}(1 + |h|).$$

For example, dominate the real values on the grid $D^{-1}\mathbb{Z}$ by $C(1 + |n||g|/(D|h|))^{-2}$, and bound the resulting series by its first term plus its integral. Multiplying by $|f(h)|$ is integrable by SWQ3. Hence

$$f * E\xi = E\eta \in V. \quad (\text{SWQ14})$$

Commutativity gives the ideal property in both orders. This proves the actual ideal assertion needed for the quotient, including the source-domain check in Lemma `varthetaH1V`.

Let

$$W = \overline{V} \subset B, \quad Q = B/W, \quad (\text{SWQ15})$$

where closure is in the topology stated before SWQ3. Continuity of convolution by each fixed test function, established in SWQ4, makes W a convolution ideal. Continuity and involutivity of sharp in SWQ5, together with SWQ9–10, give $W^\sharp = W$. Thus $[f] * [g] = [f * g]$ and $[f]^\sharp = [f^\sharp]$ are well-defined on Q . Notice that SWQ13 is not an exchange of $\sum_{q \neq 0}$ with the degree integral of $E\xi$. It does not assert $\int_C |g| E\xi(g) d^\times g = 0$. The author's `fubini1-nothappen` calculation concerns precisely that different, generally invalid exchange.

3.4 The full-label quotient and the different Bourne quotient

Let L be any bounded distributive lattice. All constructions below hold without finiteness; to require $e \neq \tau$ take L nontrivial. Define the original constrained carrier and operations

$$\begin{aligned} G_L(B) &= \{(0, \lambda) : \lambda \in L\} \cup (B \times \{1_L\}), \\ (f, \lambda) + (g, \mu) &= (f + g, \lambda \vee \mu), \quad (f, \lambda) * (g, \mu) = (f * g, \lambda \wedge \mu), \\ (f, \lambda)^\sharp &= (f^\sharp, \lambda), \quad \tau = (0, 0_L), \quad e = (0, 1_L). \end{aligned} \quad (\text{SWQ16})$$

This is a commutative nonunital convolution semiring with sharp. Carrier closure follows because any nonzero output amplitude requires all relevant nonzero input amplitudes to have top label. For the product this uses that convolution with a zero function is zero; it does not assume that B has no zero divisors. Associativity and distributivity combine those of convolution with the lattice laws. Sharp is an involution preserving labels and reversing convolution. The complex scalar action has the separate semimodule formula $(c, \alpha)(f, \lambda) = (cf, \alpha \wedge \lambda)$ for $(c, \alpha) \in G_L(\mathbb{C})$.

The exact same-label quotient is

$$(f, \lambda) \sim_W (g, \mu) \iff \lambda = \mu \text{ and } f - g \in W, \quad G_L(B)/\sim_W \simeq G_L(Q). \quad (\text{SWQ17})$$

To prove this, use the map $\pi_L(f, \lambda) = ([f], \lambda)$. It respects addition, convolution, sharp, and the scalar action by the ideal and involution properties just proved. It is onto: a nonzero quotient amplitude has top label and lifts to a representative with top label; a zero amplitude with any label lifts to $(0, \lambda)$. Its fibres are exactly the displayed relation. Alternatively, adding a third element preserves equality of labels and preserves the amplitude difference modulo W ; multiplication sends that difference to an element of W ; sharp sends it into W as well. These direct checks prove the required congruence and the quotient assertion.

In particular every $(0, \lambda)$ survives, and e and τ remain distinct for nontrivial L . A nonzero $w \in W$ with top label maps to e , which differs from τ when L is nontrivial. For nontrivial L , the inverse image of the absorbing additive zero τ under π_L is only $\{\tau\}$, even though π_L can identify different top-labelled amplitudes. Thus the preimage of a single zero does not specify this semiring congruence. For the one-element lattice instead $G_L(B) = B$, and that inverse image is the ordinary amplitude kernel W .

The ordinary Bourne congruence by the ideal $I = G_L(W)$ is different:

$$x \equiv_I y \iff \exists a, b \in I : x + a = y + b, \quad (f, \lambda) \equiv_I (g, \mu) \iff f - g \in W. \quad (\text{SWQ18})$$

The forward implication follows from the amplitude equation. Conversely, if $f - g \in W$, choose $a = (g - f, 1_L)$ and $b = (0, 1_L)$. Both belong to I ; the two amplitudes become g and both labels become 1_L , proving equality. Consequently

$$G_L(B)/\equiv_I \simeq Q, \quad G_L(B)/\sim_W \longrightarrow Q, \quad ([f], \lambda) \longmapsto [f]. \quad (\text{SWQ19})$$

The Bourne quotient collapses the entire labelled-zero fibre, so it identifies e and τ . More exactly it is the same-label quotient followed by the congruence generated by $e = \tau$: multiplication of this equality by $(0, \lambda)$ makes $(0, \lambda) = \tau$ for every λ . After these identifications the carrier and operations are exactly those of Q . No Boolean character or selected support point has been used.

3.5 Every numerical additive trace and the full-labelled receiver

Let A be any abelian group. Every additive map $t : G_L(B) \rightarrow A$ kills every labelled zero: its idempotence gives $t(0, \lambda) + t(0, \lambda) = t(0, \lambda)$, hence $t(0, \lambda) = 0$. Define $\ell(f) = t(f, 1_L)$. Then ℓ is additive on B , since addition of two top-labelled elements stays top-labelled and $\ell(0) = t(e) = 0$. Every nonzero amplitude already has top label, so

$$t(f, \lambda) = \ell(f), \quad \text{Hom}_+(G_L(B), A) \simeq \text{Hom}_+(B, A). \quad (\text{SWQ20})$$

Conversely composition of any additive ℓ with amplitude projection is additive, proving both directions. This includes all numerical additive traces into $\mathbb{C}$, not only continuous or complex-linear ones. Cyclicity adds no restriction here because convolution is commutative. Such a map descends through SWQ17 exactly when $\ell(W) = 0$: necessity follows by comparing $(w, 1_L)$ with e , and sufficiency follows from the displayed difference condition. The same classification on $G_L(Q)$ is $\text{Hom}_+(G_L(Q), A) = \text{Hom}_+(Q, A)$. Thus numerical additivity itself removes zero-label distinctions; it is not a proof that those distinctions were absent from the source object.

Retain the original numerical cohomological trace

$$T(f) = \text{Tr}(\vartheta_m(f) \mid H^1). \quad (\text{SWQ21})$$

Here $\vartheta_m(f)$ is the original convolution action on the source's quotient realization, with its prescribed closure. Its trace-class meaning is the cited Theorem `varthetatagTrH1`, with the trace formula `TraceformulaThm`; this chapter does not replace that analytic theorem by a finite-dimensional trace assumption. Its vanishing on W can be checked on the actual quotient action: for $w \in W$ and $b \in B$, $w * b \in W$, hence $\vartheta_m(w)[b] = 0$. The operator is zero, so

$$T(W) = 0, \quad \bar{T} : Q \rightarrow \mathbb{C}, \quad \bar{T}([f]) = T(f). \quad (\text{SWQ22})$$

This does not require an unproved exchange of an infinite spectral sum with a limit of representatives: the operator itself vanishes.

The original trace and Weil-pairing receivers retaining the labels are

$$\begin{aligned} T_L : G_L(Q) &\longrightarrow G_L(\mathbb{C}), & T_L([f], \lambda) &= (T(f), \lambda), \\ \mathcal{W}_L(([f], \lambda), ([g], \mu)) &= (T(f * g^\sharp), \lambda \wedge \mu). \end{aligned} \quad (\text{SWQ23})$$

The first map is well-defined by SWQ22. It respects addition and the declared $G_L(\mathbb{C})$ scalar action; it is a trace receiver, not a claimed multiplicative character. For the second, replace f by $f + w$ and g by $g + v$, with $w, v \in W$. Its change of amplitude is

$$T(w * g^\sharp + f * v^\sharp + w * v^\sharp) = 0. \quad (\text{SWQ24})$$

Every summand belongs to W by SWQ15. This proves representative independence while retaining each term of the actual change. If one input quotient amplitude is zero, the output amplitude is zero. Thus every nonzero output amplitude has top label and the receiver belongs to the declared constrained carrier.

The receiver is additive in both variables and complex-linear in the first, conjugate-linear in the second, with the exact lattice factors. For example, if $a = (c, \alpha)$, then

$$\mathcal{W}_L(ax, y) = a\mathcal{W}_L(x, y), \quad \mathcal{W}_L(x, ay) = \bar{a}\mathcal{W}_L(x, y), \quad \bar{a} = (\bar{c}, \alpha). \quad (\text{SWQ25})$$

The amplitudes follow from convolution and sharp. For additivity, the label equality is $(\lambda \vee \lambda') \wedge \mu = (\lambda \wedge \mu) \vee (\lambda' \wedge \mu)$, and likewise in the second variable. These prove the assertions directly. In particular $\mathcal{W}_L((0, \lambda), ([g], \mu)) = (0, \lambda \wedge \mu)$. Even a zero numerical amplitude therefore retains its exact support in SWQ23. Projecting to $\mathbb{C}$ forgets that support by SWQ20.

3.6 The complete original trace terms of this receiver

For the trivial unit character write $Mf(s) = \int_C f(u)|u|^s d^\times u$. Every such integral converges absolutely on the full complex plane by SWQ3, locally uniformly with every derivative. Haar inversion gives

$$M(f^\sharp)(s) = \overline{Mf(1-\bar{s})}, \quad M(f * g^\sharp)(s) = Mf(s) \overline{Mg(1-\bar{s})}. \quad (\text{SWQ26})$$

For the second identity the weighted double integral is finite, so multiplicative Fubini is valid. At the identity one has instead the exact weighted inner-product expression

$$(f * g^\sharp)(1) = \int_C f(u) \overline{g(u)} |u| d^\times u. \quad (\text{SWQ27})$$

In particular $M(f^\sharp)(1) = \overline{Mf(0)}$. The conjugation is necessary for complex tests; it is missing in the author's displayed `codegcorrZf` equality if that equality is read for unrestricted complex-valued f . No real-valued restriction is imposed here.

Put $h = f * g^\sharp$. The source's full trace formula therefore gives the following amplitude for SWQ23:

$$\begin{aligned} T(h) &= Mf(0) \overline{Mg(1)} + Mf(1) \overline{Mg(0)} \\ &\quad - (\log |a|) \int_C f(u) \overline{g(u)} |u| d^\times u \\ &\quad - \sum_{v \leq \infty} \int'_{(\mathbb{Q}_v^\times, e_v)} \frac{h(u^{-1})}{|1-u|_v} d_v^\times u. \end{aligned} \quad (\text{SWQ28})$$

Here u denotes the idele class of its original local embedding, a is the differential idele relating the global additive character to the specified local characters, and $d_v^\times u$ and the primed distributions are exactly those of `TraceformulaH1` and `princvalrhobeta`. In the canonical rational character used in SWQ6 one has $a_v = 1$ at every place, so the displayed differential term evaluates with $\log |a| = \log 1 = 0$; it has not been absorbed into another term. The prime sum includes infinity and every finite prime.

The prime notation is essential. In the author's definition a distribution ϱ_{e_v} extends multiplicative measure across zero with Fourier transform satisfying $\widehat{\varrho}_{e_v}(1) = 0$, and the local term is

$$\left\langle \varrho_{e_v}, \lambda \mapsto \frac{h((\lambda+1)^{-1})}{|\lambda+1|_v} \right\rangle.$$

For compactly supported tests this is the stated original definition; for the full space B , SWQ28 uses the extension on that test space in the cited trace theorem. It is not an assertion that the ordinary integral through $u = 1$ converges. No replacement of f by $|\cdot|^{1/2}f$, and no completed zeta function, is introduced into SWQ23–28.

Exact scope and analytic limits. The closure in SWQ15 is the declared weighted Bruhat closure, not an unspecified Hilbert, distributional, or pointwise closure. The quotient, ideal, and sharp statements above are proved in this domain. The scalar trace-class realization and its regularized explicit formula retain the analytic hypotheses of the original source theorems. SWQ23 does not supply a new proof of those input theorems or a positivity result. It supplies a fully specified same-label receiving object and proves how its operations and every trace term descend. Its numerical values are exactly the original ones: every such value is attained on top-labelled inputs, and every labelled input has the displayed numerical amplitude. The support data remain in $G_L(Q)$ and in $G_L(\mathbb{C})$; an additive numerical observation necessarily forgets them by SWQ20. The Bourne quotient performs that forgetting already in its source, whereas SWQ17 retains $e \neq \tau$ for every nontrivial L .

4 Actual trivial correspondences, spectral jets, and supported nilpotence

Source and scope. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author file `Yuri70v2.tex`, define the restriction image in `rangecVdef`, the induced convolution action in `varthetaconvol`, and prove its trace vanishing in `vanishV`. Their `degmodifyV` uses the particular Schwartz function below, stating its transform only up to a constant factor. The factor is retained here. The interpolation and support calculations are the following programme derivation, not a claim that these proofs occur verbatim in that source.

Let $k = (k_p)_p$ have finite support in $\bigoplus_p \mathbb{Z}$, let $a > 0$, and retain

$$N(k) = \prod_p p^{k_p}, \quad \eta(x) = \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, \quad \xi_{k,a}(x) = \eta(x_\infty/a) \prod_p \mathbf{1}_{p^{k_p} \mathbb{Z}_p}(x_p). \quad (\text{WJI1})$$

The additive Haar measures have $\text{vol}(\mathbb{Z}_p) = 1$; on $C_{\mathbb{Q}} = \widehat{\mathbb{Z}}^* \times \mathbb{R}_+^*$ retain Haar mass one on $\widehat{\mathbb{Z}}^*$ and dy/y . Here $\xi_{k,a}(0) = 0$ and its additive integral is zero: the two real Gaussian moments are $\pi^2 \cdot 3/(4\pi^2)$ and $(3\pi/2) \cdot 1/(2\pi)$, and their difference is zero. The finite Haar factor is $N(k)^{-1}$ and the real change of variable contributes a ; neither factor has been discarded.

For $u \in \widehat{\mathbb{Z}}^*$ put

$$f_b(u, y) = E\xi_{k,a}(u, y) = \sum_{q \in \mathbb{Q}^*} \xi_{k,a}(q(u, y)) = 2 \sum_{n \geq 1} \eta(ny/b), \quad b = a/N(k). \quad (\text{WJI2})$$

Indeed the finite factors require $q \in N(k)\mathbb{Z}$, and η is even. This proves the factor 2 and the exact relation of b to every original prime valuation and real width. The function is in the restriction image $\mathcal{V}$ because $\xi_{k,a}$ satisfies both original vanishing conditions.

For completeness, the real Fourier transform of η is η . One differentiates the Gaussian identity $\mathcal{F}(e^{-\pi x^2}) = e^{-\pi v^2}$ using $\mathcal{F}(xf) = -(2\pi i)^{-1} \partial_v \mathcal{F}f$. The transforms of $x^2 e^{-\pi x^2}$ and $x^4 e^{-\pi x^2}$ are respectively $(1/(2\pi) - v^2) e^{-\pi v^2}$ and $(v^4 - 3v^2/\pi + 3/(4\pi^2)) e^{-\pi v^2}$. Substitution of both terms of η proves the assertion. Poisson summation therefore gives

$$f_b(y) = \frac{b}{y} f_{1/b}(1/y). \quad (\text{WJI3})$$

There is no omitted zero summand: $\eta(0) = \widehat{\eta}(0) = 0$. Both f_b and every logarithmic derivative decay faster than every power at infinity, by the Gaussian series. WJI3 proves the same at zero. Thus f_b lies in the exact radial strong Schwartz algebra used in the source, with no compact-support replacement.

The full Mellin calculation, first for $\Re s > 1$, is

$$\mathcal{M}f_b(s) := \int_0^\infty f_b(y) y^s \frac{dy}{y} = b^s C(s), \quad C(s) = \pi^{-s/2} \zeta(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right]. \quad (\text{WJI4})$$

Absolute convergence for the separate Gaussian terms follows from $\sum n^{-\Re s} < \infty$. For each term substitute $v = \pi(ny/b)^2$; its factor $1/2$ cancels the displayed factor 2 in WJI2. This yields exactly the two Gamma summands. WJI3 gives an entire Mellin transform: for any compact set of s , its derivatives introduce powers of $\log y$ dominated at both ends by the proved estimates.

The Gamma recurrence proves the comparison, without replacing the original formula WJI4:

$$C(s) = d(s) \zeta(s), \quad d(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2). \quad (\text{WJI5})$$

At zero use $\zeta(0) = -1/2$ and the bracket $\Gamma(2) - (3/2)\Gamma(1) = -1/2$ in WJI4. At one the bracket is $(s-1)\Gamma(s/2+1)/2$ and ζ has residue one. Consequently

$$C(0) = \frac{1}{4}, \quad C(1) = \frac{1}{4}, \quad \mathcal{M}f_b(0) = \frac{1}{4}, \quad \mathcal{M}f_b(1) = \frac{b}{4}. \quad (\text{WJI6})$$

The original zeta continuation with both endpoint terms is proved in ZTC9–10; the values and residue used here follow from that formula. No zero or pole is assigned a value by multiplying undefined separate point values.

4.1 The exact finite-jet image

Fix distinct complex points $s_1, \dots, s_\ell$, positive integers $N_1, \dots, N_\ell$, and use the original local coordinates $h_i = s - s_i$. Let

$$J = \bigoplus_{i=1}^{\ell} \mathbb{C}[h_i]/(h_i^{N_i}), \quad m_i = \text{ord}_{s_i} C, \quad a_i = \min(m_i, N_i), \quad r_i = N_i - a_i, \quad M = \sum_i r_i. \quad (\text{WJI7})$$

The nonnegative orders m_i are defined by the entire function WJI4; they are finite since $C(0) = 1/4$. Define the actual Mellin-jet map on finite linear combinations of the functions WJI2 by

$$\mathcal{J} \left(\sum_j c_j f_{b_j} \right) = \left[\sum_j c_j b_j^s C(s) \right]_{h_i^{N_i}, i}.$$

Every such jet lies in

$$I_J = \bigoplus_i h_i^{a_i} \mathbb{C}[h_i]/(h_i^{N_i}). \quad (\text{WJI8})$$

We now prove that every element of WJI8 occurs and give its inverse construction using M specified positive widths.

Discard indices with $r_i = 0$ only from the interpolation matrix; their zero image and quotient remain in WJI7–8. If $M = 0$, the image is zero and the empty linear combination suffices. Otherwise choose, on the retained indices,

$$\delta = \frac{\pi}{1 + \max_{i < j} |\Im(s_i - s_j)|}, \quad z_i = e^{\delta s_i}, \quad b_j = e^{j\delta} \quad (0 \leq j < M), \quad (\text{WJI9})$$

with maximum zero when there is only one index. The z_i are distinct. Equality of two exponentials would require $\delta(s_i - s_j) = 2\pi in$. The chosen bound makes the imaginary part smaller than π in absolute value, forcing $n = 0$ and then $s_i = s_j$, a contradiction.

The derivative matrix for the functions $e^{j\delta s}$ is

$$V_{(i,q),j} = (j\delta)^q z_i^j, \quad 0 \leq q < r_i, \quad \det V = \left[\prod_i (\delta z_i)^{r_i(r_i-1)/2} \prod_{q=0}^{r_i-1} q! \right] \prod_{i < j} (z_j - z_i)^{r_i r_j}. \quad (\text{WJI10})$$

Every factorial in this formula refers to ordinary derivatives, not Taylor coefficients. To verify it, first use polynomial derivative evaluations $P^{(q)}(z_i)$. Their confluent Vandermonde determinant equals the ordinary Vandermonde limit: replace each cluster z_i by $z_i + \epsilon w_{i,j}$, divide by its within-cluster Vandermonde, and expand each row in its Taylor series. The surviving rows are $P^{(q)}(z_i)/q!$; between-cluster factors tend to $(z_j - z_i)^{r_i r_j}$. Multiplying the rows by $q!$ gives the displayed factorial product. Finally $\partial_s = \delta z \partial_z$ has leading coefficient $(\delta z_i)^q$ in its q th power and only lower-order derivatives besides. This triangular row change proves WJI10. All its factors are nonzero.

For an explicit inverse, let a desired jet in WJI8 have at site i the representative $g_i(h_i)$ divisible by $h_i^{m_i}$. Write $C(s_i + h) = h^{m_i} u_i(h)$, $u_i(0) \neq 0$. Division gives the unique required jet

$$w_i(h) = \frac{g_i(h)}{h^{m_i} u_i(h)} \pmod{h^{r_i}}. \quad (\text{WJI11})$$

The ambiguity in a representative of g_i is divisible by h^{N_i} , so the ambiguity after division is divisible by h^{r_i} . Put

$$Q_i(z) = \prod_{\nu \neq i} (z - z_\nu)^{r_\nu}, \quad \tilde{w}_i(z) = w_i(\delta^{-1} \log(z/z_i)) \pmod{(z - z_i)^{r_i}},$$

where the local logarithm has value zero at z_i . The polynomial

$$P(z) = \sum_i Q_i(z) \operatorname{Tay}_{z_i, < r_i} \left(\frac{\tilde{w}_i(z)}{Q_i(z)} \right) = \sum_{j=0}^{M-1} c_j z^j \quad (\text{WJI12})$$

has degree below M . Its i th summand has the prescribed jet at z_i ; every other summand is divisible by $(z - z_i)^{r_i}$. Hence $C(s)P(e^{\delta s})$ has the desired original s -jets. The actual correspondence $\sum_{j=0}^{M-1} c_j f_{b_j}$ realizes them. This proves

$$\operatorname{im} \mathcal{J} = I_J, \quad \operatorname{rank} \mathcal{J} = \sum_i (N_i - a_i), \quad J/I_J = \bigoplus_i \mathbb{C}[h_i]/(h_i^{a_i}). \quad (\text{WJI13})$$

The equality on the right is the explicit residue map; its kernel is exactly WJI8. A summand with $a_i = 0$ is the zero vector space, not a omitted original point or support label.

4.2 Endpoints, original divisors and the full nilpotent

At $s_i = 0$ or 1 , WJI6 gives $a_i = 0$ for every N_i . Thus all finite jets at both endpoints can be changed independently by actual elements of the restriction image. For only the two values and two widths $b_1 \neq b_2$ the explicit equations are

$$\frac{c_1 + c_2}{4} = A_0, \quad \frac{b_1 c_1 + b_2 c_2}{4} = A_1, \quad c_1 = \frac{4(A_1 - b_2 A_0)}{b_1 - b_2}, \quad c_2 = \frac{4(b_1 A_0 - A_1)}{b_1 - b_2}. \quad (\text{WJI14})$$

Adding this correspondence changes the two Mellin values by A_0, A_1 and leaves its class modulo $\mathcal{V}$ unchanged. It leaves every trace pairing against that class unchanged, by the source's `vanishV`; the full-labelled meaning of that assertion is SWQ, not an identification of e and τ .

For a nontrivial original zeta zero ρ in $0 < \Re \rho < 1$, the factor d in WJI5 is a holomorphic unit. Hence $m_\rho = \operatorname{ord}_\rho \zeta$ and the quotient in WJI13 retains the complete truncated original primary block. At $s = -2n$, however, d has a simple pole and C is a unit. The corresponding WJI13 quotient is zero, but the original simple zero of ζ remains in $d^{-1}\mathcal{O} = h\mathcal{O} \subset \mathcal{O}$ and the length-one quotient $\mathcal{O}/h\mathcal{O}$. At $s = 1$, $d^{-1}\mathcal{O} = h^{-1}\mathcal{O}$ and $h^{-1}\mathcal{O}/\mathcal{O}$ retains the original pole with the opposite divisor sign. The embedded comparison and every factor of WJI5 are part of this calculation; WJI13 is not asserted to be the complete original-zeta divisor by itself. The germ proofs, including the module-fibre isomorphism and all exceptional-point signs, are ZTC17–17a.

The original scaling action $S_r f(y) = f(r^{-1}y)$, $r > 0$, sends f_b to f_{rb} and gives $\mathcal{M}(S_r f)(s) = r^s \mathcal{M}f(s)$ by substitution. It preserves WJI8. On a nonzero local quotient in WJI13 write N_i for multiplication by h_i ; the symbol N_i remains the integer cutoff of WJI7. The exact action is

$$N_i[v] = [h_i v], \quad N_i^{a_i} = 0, \quad S_r = r^{s_i} \sum_{q=0}^{a_i-1} \frac{(\log r)^q}{q!} N_i^q. \quad (\text{WJI15})$$

This is the Taylor expansion of $r^{s_i+h_i}$ in the specified quotient, with every factorial and eigenvalue retained. The vector [1] shows that its nilpotent length is exactly a_i : for $a_i \geq 1$, $[h_i^{a_i-1}] \neq 0$. No semisimplification of this primary block has been taken.

Let L be the original arbitrary nontrivial bounded distributive lattice. For any vector space V use exactly

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad (v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu). \quad (\text{WJI16})$$

The $G_L(\mathbb{C})$ action multiplies amplitudes and meets labels. Every linear map T has the lift $T^\uparrow(v, \lambda) = (Tv, \lambda)$, which preserves addition and that action. The jet quotient has exact same-label fibres

$$(v, \lambda) \sim (w, \mu) \iff \lambda = \mu \text{ and } v - w \in I_J. \quad (\text{WJI17})$$

Addition and scalar action preserve this relation because I_J is a vector subspace. The map to $G_L(J/I_J)$ is surjective: use any lift of a nonzero amplitude with label 1_L , and use $(0, \lambda)$ for each zero-labelled output. Its fibres are precisely WJI17, proving the quotient identification with all original labels retained.

For each nonzero block its lifted nilpotent satisfies

$$(\mathbf{N}_i^\dagger)^{a_i}(v, \lambda) = (0, \lambda) = e \cdot (v, \lambda), \quad e = (0, 1_L), \quad \tau = (0, 0_L). \quad (\text{WJI18})$$

This map is not the constant- τ map. On the zero vector block ($a_i = 0$), $G_L(0) = L$ and the scaling action is the identity on that lattice; an empty polynomial sum must not be read as the constant- τ map.

These are common-label modules for the entire indicated vector spaces. They do not introduce independent labels on each jet coordinate. In the actual base localization of ULC, at a support prime $J_L \subset L$, the element e is invertible, so every amplitude is killed and the coefficient stalk is L_{J_L} . Explicitly $(v, \lambda)/e = (0, \lambda)/e$, because multiplication by e makes their numerators equal. Each lifted linear map then acts as the identity on L_{J_L} , and WJI18 becomes the identity there. On the arithmetic coefficient modules, WJI15 and WJI18 retain the primary block and its supported-zero power separately. This is the exact receiving map; it asserts no new positivity of the original Weil form.

5 The exact Hurwitz boundary of the Connes–Consani–Marcolli test space

Sources and scope. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source `Yuri70v2.tex`, equation `SCK`, Proposition `Trrhocyclmap`, equations `TrMbf0` and `rangeCVdef`, and Lemma `degmodifyV`, provide the actual strong test space, restriction map and Gaussian-polynomial generator used here. The source’s wording “up to normalization” in the proof of `degmodifyV` is not used as a formula: every factor of its original generator is retained and calculated below. The original Hurwitz family is the programme’s $F_t(s) = \zeta(s, 1 + t)$, as recorded in HTB1–4 of `HURWITZ_THETA_FLOW_BRIDGE.tex`; a direct continuation proof is included here. Only the cited parts of the CCM source were read for this independent derivation. The results concern the actual test space and its explicit boundary, without an RH claim.

5.1 Objects, original restriction map, and all constants

Write $E = y\partial_y$, $a = 1 + t > 0$, and $b > 0$. Define

$$\begin{aligned}\eta(x) &= \pi x^2(\pi x^2 - \tfrac{3}{2})e^{-\pi x^2}, \\ g(x) &= 2\eta(x) = (2\pi^2 x^4 - 3\pi x^2)e^{-\pi x^2}, \\ \Psi_{t,b}(y) &= \sum_{n=0}^{\infty} g((n+1+t)y/b), \quad y > 0, \\ \mathcal{B} &= \bigcap_{\beta \in \mathbb{R}} y^\beta \mathcal{S}(\mathbb{R}_+^*). \end{aligned} \tag{HCB1}$$

Here $\mathcal{S}(\mathbb{R}_+^*)$ is Schwartz space in the coordinate $\log y$, and $\mathcal{B}$ is the radial part of CCM’s $\mathbf{S}(C_{\mathbb{Q}})$. Equivalently, $u \in \mathcal{B}$ exactly when u is smooth and

$$\sup_{y>0} (y^N + y^{-N})|E^r u(y)| < \infty \quad \text{for every pair of integers } N, r \geq 0. \tag{HCB2}$$

Indeed, the defining weighted Schwartz seminorms become $\sup_x (1 + |x|)^j |\partial_x^r (e^{-\beta x} u(e^x))|$. HCB2 bounds each of their finitely many Leibniz summands by choosing $N > |\beta| + 1$, since $(1 + |x|)^j e^{-|x|}$ is bounded. Conversely, the weighted Schwartz seminorms with $\beta = N$ and $\beta = -N$, and induction in r on the Leibniz formula, give HCB2. No condition at either endpoint is discarded.

Take additive Haar measure with $\text{vol}(\widehat{\mathbb{Z}}) = 1$, and put

$$\xi_b = \mathbf{1}_{\widehat{\mathbb{Z}}} \otimes \eta(\cdot/b), \quad f_b = \text{Tr } \rho(\xi_b). \tag{HCB3}$$

The Gaussian moments are $\int_{\mathbb{R}} x^2 e^{-\pi x^2} dx = 1/(2\pi)$ and $\int_{\mathbb{R}} x^4 e^{-\pi x^2} dx = 3/(4\pi^2)$, obtained by twice differentiating $\int e^{-\pi c x^2} dx = c^{-1/2}$ for $c > 0$. For this last identity, the square of the positive integral at $c = 1$ equals $\int_0^\infty 2\pi r e^{-\pi r^2} dr = 1$ by polar coordinates; scaling gives the stated value for every $c > 0$. Consequently $\eta(0) = 0$ and $\int \eta = 3/4 - 3/4 = 0$. Thus ξ_b is in the actual source $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \ker \varepsilon_0 \cap \ker \varepsilon_1$. CCM’s restriction formula, on $(u, y) \in \widehat{\mathbb{Z}}^\times \times \mathbb{R}_+^*$, gives

$$f_b(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \eta(ny/b) = \Psi_{0,b}(y), \quad \Psi_{-1/2,b} = f_{2b} - f_b = \text{Tr } \rho(\xi_{2b} - \xi_b). \tag{HCB4}$$

The first equality uses that $ku \in \widehat{\mathbb{Z}}$ for $k \in \mathbb{Q}$ precisely when $k \in \mathbb{Z}$. The second identity partitions the positive integers into their odd and even members. These are actual restriction-image elements, not merely elements with the same Mellin zeros.

5.2 An exact differentiated Euler–Maclaurin formula

Define Bernoulli polynomials, including every coefficient, by

$$\frac{ze^{az}}{e^z - 1} = \sum_{j=0}^{\infty} B_j(a) \frac{z^j}{j!}, \quad |z| < 2\pi, \quad B_j = B_j(0). \quad (\text{HCB5})$$

Multiplication of these power series proves $B_j(a+1) - B_j(a) = ja^{j-1}$, $B'_j(a) = jB_{j-1}(a)$ and $B_j(1-a) = (-1)^j B_j(a)$. For $j \geq 2$ let $\tilde{B}_j(x) = B_j(x - \lfloor x \rfloor)$, extended continuously at the integers. Also put $\tilde{B}_1(x) = x - \lfloor x \rfloor - 1/2$ off the integers and $\tilde{B}_1(x) = 0$ at the integers; its values at the jump points do not affect the distributional derivative or the integrals below. The explicitly finite constant

$$C_j = \frac{1}{j!} \sum_{\ell=0}^j \binom{j}{\ell} |B_\ell| \quad \text{satisfies} \quad \frac{|\tilde{B}_j(x)|}{j!} \leq C_j. \quad (\text{HCB6})$$

Let v and all its derivatives decrease rapidly on $[0, \infty)$, $h > 0$, $a = q + \alpha$ with $q = \lceil a \rceil - 1 \geq 0$ and $0 < \alpha \leq 1$, and take an even integer $J \geq 2$. Then

$$\begin{aligned} \sum_{n \geq 0} v((n+a)h) &= \frac{1}{h} \int_0^\infty v(x) dx - \sum_{j=1}^J \frac{B_j(a)}{j!} h^{j-1} v^{(j-1)}(0) + \mathcal{R}_J(v; a, h), \\ \mathcal{R}_J(v; a, h) &= \frac{(-1)^{J+1} h^{J-1}}{J!} \int_0^\infty \tilde{B}_J(x/h - \alpha) v^{(J)}(x) dx \\ &\quad - \sum_{n=0}^{q-1} \frac{((n+\alpha)h)^J}{(J-1)!} \int_0^1 (1-z)^{J-1} v^{(J)}(z(n+\alpha)h) dz. \end{aligned} \quad (\text{HCB7})$$

An empty finite sum is zero. To prove the formula first take $0 < \alpha < 1$. The distributional derivative of $\tilde{B}_1(x/h - \alpha)$ is $h^{-1} - \sum_{n \in \mathbb{Z}} \delta_{h(n+\alpha)}$. Integrating it against v on $[0, \infty)$, and then integrating by parts $J-1$ more times, gives the infinite-sum formula with $a = \alpha$: the endpoint identities $B_j(1-\alpha) = (-1)^j B_j(\alpha)$ give exactly the displayed signs. The rapid decrease removes all upper boundary terms. For $\alpha = 1$, take the limit from below; $J \geq 2$ makes the final periodic Bernoulli function continuous, and its uniform bound permits dominated convergence. Subtract the first q summands. Taylor's integral formula, with terms of degrees 0 through $J-1$, and the telescoped identity $B_j(a) - B_j(\alpha) = j \sum_{n < q} (n+\alpha)^{j-1}$ yield HCB7, including its second remainder term. In particular, for $0 < a \leq A$, $Q = \max(0, \lceil A \rceil - 1)$ and $0 < h \leq 1$,

$$|\mathcal{R}_J(v; a, h)| \leq C_J \|v^{(J)}\|_{L^1(0, \infty)} h^{J-1} + \frac{Q A^J}{J!} \|v^{(J)}\|_{L^\infty(0, \infty)} h^J. \quad (\text{HCB8})$$

Here are finite formulas for every constant needed when differentiating the actual generator. Set

$$\begin{aligned} P_0(x) &= 2\pi^2 x^4 - 3\pi x^2, & P_{r+1}(x) &= xP'_r(x) - 2\pi x^2 P_r(x), \\ Q_{r,0} &= P_r, & Q_{r,J+1} &= Q'_{r,J} - 2\pi x Q_{r,J}, & Q_{r,J}(x) &= \sum_j q_{r,J,j} x^j, \\ U_{r,J} &= \sum_j |q_{r,J,j}| \frac{\Gamma((j+1)/2)}{2\pi^{(j+1)/2}}, & V_{r,J} &= \sum_j |q_{r,J,j}| v_j, \\ v_0 &= 1, & v_j &= (j/(2\pi e))^{j/2} \quad (j > 0). \end{aligned} \quad (\text{HCB9})$$

Direct differentiation shows $(x\partial_x)^r g = P_r e^{-\pi x^2}$ and its J th ordinary derivative is $Q_{r,J} e^{-\pi x^2}$. Termwise integration and maximization of $x^j e^{-\pi x^2}$ give $\|((x\partial_x)^r g)^{(J)}\|_1 \leq U_{r,J}$ and its supremum at most $V_{r,J}$. These are specified finite sums, not unspecified remainder constants.

The integral of $(x\partial_x)^r g$ is zero for every $r \geq 0$: the case $r = 0$ was computed above, and integration by parts gives $\int x v'(x) dx = -\int v(x) dx$ at each step. The Taylor series is

$$g(x) = \sum_{m=1}^{\infty} \frac{(-1)^m \pi^m (2m+1)}{(m-1)!} x^{2m}, \quad ((x\partial_x)^r g)^{(2m)}(0) = (2m)^r \frac{(-1)^m \pi^m (2m+1)(2m)!}{(m-1)!}. \quad (\text{HCB10})$$

All odd derivatives and the constant term vanish. For $m = 1$ the displayed coefficient is -3π . For $m \geq 2$, it is the sum of $2\pi^2(-\pi)^{m-2}/(m-2)!$ and $-3\pi(-\pi)^{m-1}/(m-1)!$, so both original contributions have been retained in obtaining HCB10.

For integers $M \geq 1$ and $r \geq 0$, the full differentiated asymptotic coefficients and remainders are

$$c_m(t, b) = \frac{(-1)^{m+1} \pi^m B_{2m+1}(1+t)}{(m-1)! b^{2m}}, \quad E^r \Psi_{t,b}(y) = \sum_{m=1}^M (2m)^r c_m(t, b) y^{2m} + \mathcal{E}_{M,r}(t, b; y). \quad (\text{HCB11})$$

This is not merely a formal expansion. To give its uniform bound put

$$D_m(A) = \frac{\pi^m}{(m-1)!} \sum_{j=0}^{2m+1} \binom{2m+1}{j} |B_j| A^{2m+1-j}, \quad J = 2M+4, \quad (\text{HCB12})$$

$$K_{M,r,A} = (2M+2)^r D_{M+1}(A) + C_J U_{r,J} + \frac{Q A^J}{J!} V_{r,J},$$

$$|\mathcal{E}_{M,r}(t, b; y)| \leq K_{M,r,A} (y/b)^{2M+2} \quad (0 < 1+t \leq A, \quad 0 < y/b \leq 1).$$

Indeed, apply HCB7 to $v = (x\partial_x)^r g$ and $h = y/b$. The integral term and all odd Taylor terms vanish. The terms through $j = J$ give the displayed sum through $m = M+1$. HCB8 bounds its remainder by constants times h^{2M+3} and h^{2M+4} ; each is at most the same constant times h^{2M+2} on the stated domain. The omitted $m = M+1$ coefficient has modulus at most $(2M+2)^r D_{M+1}(A)$. Differentiation under the original series is valid on compact subsets of $a, b, y > 0$: each differentiated summand is a polynomial in $(n+a)y/b$ times its Gaussian, bounded by a summable sequence independent of the point in such a compact set. Thus the Euler derivatives in HCB11 are the actual derivatives.

For completeness there is also a uniform bound at the other endpoint. Write $P_r(x) = \sum_j p_{r,j} x^j$, and define

$$L_r = \sum_j |p_{r,j}| w_j, \quad w_0 = 1, \quad w_j = (j/(\pi e))^{j/2} \quad (j > 0). \quad (\text{HCB13})$$

$$|E^r \Psi_{t,b}(y)| \leq \frac{L_r}{1 - e^{-\pi/2}} e^{-\pi(1+t)^2(y/b)^2/2} \quad (y/b \geq 1).$$

This follows from $x^j e^{-\pi x^2} \leq w_j e^{-\pi x^2/2}$ and $(n+a)^2 \geq a^2 + n$ for $n \geq 0$. Consequently all derivatives have Gaussian decrease at infinity, uniformly when $a \in [a_0, A]$ and $b \in [b_0, b_1]$ with positive endpoints.

5.3 Exact strong-space membership and all Mellin residues

The first boundary coefficient is

$$c_1(t, b) = \frac{\pi t(1+t)(1+2t)}{2b^2}. \quad \Psi_{t,b} \in \mathcal{B} \iff t \in \{0, -\tfrac{1}{2}\}. \quad (\text{HCB14})$$

To prove necessity, HCB11 with $M = 1, r = 0$ gives the actual limit $\lim_{y \downarrow 0} \Psi_{t,b}(y)/y^2 = c_1(t, b)$; HCB2 forces it to be zero. Since $t > -1$, its only zeros are $0, -1/2$. For sufficiency, $B_{2m+1}(1) = B_{2m+1}(1/2) = 0$ for every $m \geq 1$. The first identity follows from the reflection identity and the unit-shift identity at $a = 0$; the second follows immediately from reflection. HCB12 with arbitrarily

large M , for every r , and HCB13 then give all HCB2 seminorms. HCB4 proves that both cases belong to the actual restriction image $\mathcal{V}$ as well.

The original Mellin identity, initially absolutely convergent for $\operatorname{Re} s > 1$, is

$$\int_0^\infty \Psi_{t,b}(y) y^{s-1} dy = b^s \pi^{-s/2} F_t(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right], \quad F_t(s) = \zeta(s, 1+t). \quad (\text{HCB15})$$

For that domain, integrate each of the two terms of g after the change of variable $x = (n+a)y/b$. The sum of the absolute integrals is bounded by $b^{\operatorname{Re} s} \sum_{n \geq 0} (n+a)^{-\operatorname{Re} s} \int_0^\infty |g(x)| x^{\operatorname{Re} s-1} dx$, which is finite. The original Gamma difference, power of π , power of b , and Hurwitz function all remain in HCB15.

Here is the continuation of the original Hurwitz factor, with the endpoint terms shown. For any integer $N \geq 0$, in $\operatorname{Re} s > -N$ it is

$$F_t(s) = \frac{1}{\Gamma(s)} \left[\int_1^\infty x^{s-1} \frac{e^{-ax}}{1-e^{-x}} dx + \int_0^1 x^{s-1} \left\{ \frac{e^{-ax}}{1-e^{-x}} - \sum_{j=0}^N \frac{B_j(1-a)}{j!} x^{j-1} \right\} dx + \sum_{j=0}^N \frac{B_j(1-a)}{j!(s+j-1)} \right]. \quad (\text{HCB16})$$

For $\operatorname{Re} s > 1$ this follows by integrating the absolutely convergent series $e^{-ax}/(1-e^{-x}) = \sum_{n \geq 0} e^{-(n+a)x}$. HCB5 shows that the bracketed Taylor remainder is $O(x^N)$ at zero. The two integrals therefore continue holomorphically on the stated domain, with compatible formulas as N increases. The Gamma recurrence gives residue $(-1)^j/j!$ at $-j$. It follows, including the zero of $1/\Gamma$ there, that

$$\operatorname{res}_{s=1} F_t(s) = 1, \quad F_t(0) = \frac{1}{2} - a, \quad F_t(-2m) = -\frac{B_{2m+1}(a)}{2m+1} \quad (m \geq 1). \quad (\text{HCB17})$$

The left side of HCB15 already converges on $\operatorname{Re} s > -2$, by HCB12–13. Its exact meromorphic continuation to $\operatorname{Re} s > -2M - 2$ is

$$\int_1^\infty \Psi_{t,b}(y) y^{s-1} dy + \int_0^1 \left[\Psi_{t,b}(y) - \sum_{m=1}^M c_m(t,b) y^{2m} \right] y^{s-1} dy + \sum_{m=1}^M \frac{c_m(t,b)}{s+2m}. \quad (\text{HCB18})$$

HCB12 gives an integrable majorant on every compact subset of that half-plane, also after any s derivative, since logarithmic factors are dominated by a smaller positive power. Thus HCB18 proves the continuation and identifies each residue with $c_m(t,b)$. They match the factors on the right side of HCB15 separately:

$$\begin{aligned} \operatorname{res}_{s=-2m} \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right] &= \frac{(-1)^m (2m+1)}{(m-1)!}, \\ \operatorname{res}_{s=-2m} \left\{ b^s \pi^{-s/2} F_t(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right] \right\} &= \frac{(-1)^{m+1} \pi^m B_{2m+1}(a)}{(m-1)! b^{2m}}. \end{aligned} \quad (\text{HCB19})$$

For $m = 1$ only the second Gamma factor has a pole, with residue $-(3/2) \cdot 2 = -3$. For $m \geq 2$ the two residues are respectively $2(-1)^{m-2}/(m-2)!$ and $-3(-1)^{m-1}/(m-1)!$. Multiplication by the unchanged value HCB17 gives HCB19; a zero coefficient means a removable singularity, not a discarded pole.

At the other exceptional points the full finite values are

$$\int_0^\infty \Psi_{t,b}(y) \frac{dy}{y} = \frac{a}{2} - \frac{1}{4}, \quad \int_0^\infty \Psi_{t,b}(y) dy = \frac{b}{4}. \quad (\text{HCB20})$$

At $s = 0$ the Gamma difference is $1 - 3/2 = -1/2$ and HCB17 applies. At $s = 1$ the Gamma difference is zero. Its derivative is $\Gamma(3/2)/2 = \sqrt{\pi}/4$: this follows by differentiating the identity

$\Gamma(s/2+2) - \frac{3}{2}\Gamma(s/2+1) = \frac{s-1}{2}\Gamma(s/2+1)$, while retaining both original terms in HCB15. Multiplying its first-order term by the original simple pole of F_t and by $b\pi^{-1/2}$ gives $b/4$. In particular the cancelled pole at 1 leaves an explicitly nonzero integral. The full original function at $t = 0$ remains $F_0(s) = \zeta(s)$ in every one of these formulas.

5.4 The full boundary sequence and mixed cancellations

Let $\mathcal{E}$ be the algebraic vector space of functions

$$\mathcal{E} = \mathcal{B} + \text{span}_{\mathbb{C}}\{\Psi_{t,b} : t > -1, b > 0\}, \quad \mathcal{C} = \text{span}_{\mathbb{C}}\{(c_m(t,b))_{m \geq 1} : t > -1, b > 0\} \subset \mathbb{C}^{\mathbb{N}}. \quad (\text{HCB21})$$

All spans mean finite sums; no completion or infinite interchange is asserted. Define the boundary map $\partial : \mathcal{E} \rightarrow \mathcal{C}$ by the unique differentiated asymptotic sequence. It has the explicit formula

$$\partial \left(v + \sum_{j=1}^k z_j \Psi_{t_j, b_j} \right) = \left(\sum_{j=1}^k z_j c_m(t_j, b_j) \right)_{m \geq 1}, \quad v \in \mathcal{B}, \quad 0 \longrightarrow \mathcal{B} \longrightarrow \mathcal{E} \xrightarrow{\partial} \mathcal{C} \longrightarrow 0. \quad (\text{HCB22})$$

Here exactness includes the analytic assertion $\ker \partial = \mathcal{B}$. To prove it, every finite sum has the HCB11 expansion with remainders bounded by the sum of the finitely many bounds HCB12. Its coefficients are unique: first take the limit after division by y^2 , then subtract that term, divide by y^4 , and continue inductively. A function in $\mathcal{B}$ has zero at every such limit, by HCB2. Thus HCB22 is independent of the presentation of the function and is linear. If all the coefficients vanish, HCB12 for arbitrarily large M bounds every Euler derivative by arbitrarily large powers of y at zero. HCB13 and the defining property of v give all the bounds at infinity. HCB2 now proves that the function itself is in $\mathcal{B}$. Surjectivity is exactly the stated definition of $\mathcal{C}$; no assertion that it is all sequence space is made.

For every $0 < a < 1$, a genuine mixed cancellation is

$$\Psi_{a-1,b}(y) + \Psi_{-a,b}(y) = \sum_{n \in \mathbb{Z}} g((n+a)y/b) \in \mathcal{B}, \quad c_m(a-1, b) + c_m(-a, b) = 0 \quad (m \geq 1). \quad (\text{HCB23})$$

The function identity uses the evenness of g to reindex the negative integers; every sum converges absolutely locally with all derivatives. The coefficient identity is the odd-degree Bernoulli reflection identity from HCB5. HCB22 then proves actual strong-space membership, including every derivative. Thus HCB14 classifies the individual shifted generators, not arbitrary mixed combinations.

There is also an exact small-endpoint formula for HCB23. Under the Fourier convention $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{-2\pi i x \xi} dx$, one has $\widehat{g} = g$. To verify all constants, differentiate the Gaussian transform $\widehat{e^{-\pi x^2}} = e^{-\pi \xi^2}$ to obtain

$$\widehat{x^2 e^{-\pi x^2}} = (1/(2\pi) - \xi^2) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}} = (\xi^4 - 3\xi^2/\pi + 3/(4\pi^2)) e^{-\pi \xi^2}.$$

The Gaussian transform identity itself follows by differentiating its absolutely convergent integral: integration by parts gives $I'(\xi) = -2\pi \xi I(\xi)$ and the preceding Gaussian integral gives $I(0) = 1$, which determine $I(\xi) = e^{-\pi \xi^2}$. The two constant terms cancel after multiplication by $2\pi^2$ and -3π , and the remaining expression is precisely g . Periodizing a Schwartz function and integrating each Fourier coefficient over $[0, 1]$ proves Poisson summation in this case; absolute convergence of all differentiated Fourier series justifies evaluation at every a . It gives

$$\Psi_{a-1,b}(y) + \Psi_{-a,b}(y) = \frac{2b}{y} \sum_{k=1}^{\infty} \cos(2\pi k a) g(kb/y), \quad 0 < a < 1. \quad (\text{HCB24})$$

The $k = 0$ term is exactly zero because $g(0) = 0$. This equality retains every Fourier mode; it does not replace the original left-hand functions by an asymptotic approximation.

Some mixed functions have an explicit receiver in the actual CCM restriction image. If $q \geq 2$ is an integer, partitioning the positive integers into residue classes modulo q gives

$$\sum_{r=1}^{q-1} \Psi_{r/q-1,b} = f_{qb} - f_b = \text{Tr } \rho(\xi_{qb} - \xi_b). \quad (\text{HCB25})$$

The original first coefficient vanishes in this sum, and HCB22 shows that every coefficient vanishes, as can also be checked by the Bernoulli multiplication identity. Three individual reflected pairs have the following more precise formulas:

$$\begin{aligned} \Psi_{-2/3,b} + \Psi_{-1/3,b} &= f_{3b} - f_b, \\ \Psi_{-3/4,b} + \Psi_{-1/4,b} &= f_{4b} - f_{2b}, \\ \Psi_{-5/6,b} + \Psi_{-1/6,b} &= f_{6b} - f_{3b} - f_{2b} + f_b. \end{aligned} \quad (\text{HCB26})$$

For denominator 3 the two residue classes exclude multiples of 3. For denominator 4 they are exactly the odd integers at scale $4b$. For denominator 6 they exclude multiples of 2 and 3 at scale $6b$; inclusion-exclusion gives the four terms, with the displayed signs. HCB3 maps the respective combinations $\xi_{3b} - \xi_b$, $\xi_{4b} - \xi_{2b}$ and $\xi_{6b} - \xi_{3b} - \xi_{2b} + \xi_b$ to these functions. Each source has both endpoint functionals zero. For a general reflected pair HCB23 proves membership in $\mathcal{B}$; no assertion of membership in $\mathcal{V}$ follows from that calculation alone.

5.5 The retained boundary pair and every support label

Retain the original defect function as a function, not solely its sequence:

$$R_{t,b} = \Psi_{0,b} - \Psi_{t,b}, \quad (\partial \Psi_{t,b}, \partial R_{t,b}) = ((c_m(t,b))_{m \geq 1}, (-c_m(t,b))_{m \geq 1}). \quad (\text{HCB27})$$

No coefficient has been removed: the cancellation is a relation between two retained coordinates. Define the actual space of pairs and its invertible map by

$$\begin{aligned} \mathcal{D} &= \{(u, v) \in \mathcal{E} \times \mathcal{E} : u + v \in \mathcal{B}\}, \\ \mathcal{E} \oplus \mathcal{B} &\xrightarrow{A} \mathcal{D}, \quad A(u, w) = (u, w - u), \quad A^{-1}(u, v) = (u, u + v). \end{aligned} \quad (\text{HCB28})$$

Both displayed maps are linear, well defined, and their two compositions are identities by direct addition. The original pair is $A(\Psi_{t,b}, f_b)$. Projection of this pair onto its sum is f_b for every t and has kernel $\{(u, -u) : u \in \mathcal{E}\}$, so that projection loses the actual deformation. By contrast, the two-coordinate map HCB28 retains it exactly. There is also the exact quotient map

$$\mathcal{D} \longrightarrow \mathcal{B} \oplus \mathcal{C}, \quad (u, v) \longmapsto (u + v, \partial u), \quad \ker = \{(u, -u) : u \in \mathcal{B}\}. \quad (\text{HCB29})$$

To see surjectivity, choose u with any prescribed sequence in $\mathcal{C}$ and set $v = w - u$ for prescribed $w \in \mathcal{B}$. Its kernel is exactly as stated by HCB22. Thus this quotient retains every boundary coefficient but does not purport to retain the entire original function modulo no relation.

For completeness, let L be any nontrivial bounded distributive lattice and use the original full carrier

$$\begin{aligned} G_L(W) &= \{(0, \lambda) : \lambda \in L\} \cup \{(w, 1_L) : w \in W\}, \\ (w, \lambda) + (v, \mu) &= (w + v, \lambda \vee \mu), \\ (z, \nu)(w, \lambda) &= (zw, \nu \wedge \lambda). \end{aligned} \quad (\text{HCB30})$$

The scalars are $G_L(\mathcal{C})$; the last formula is well defined because a nonzero scalar or vector amplitude has top label. For any complex-linear map $T : W \rightarrow W'$, set $G_L(T)(w, \lambda) = (Tw, \lambda)$. The two formulas HCB30 prove directly that this is a $G_L(\mathcal{C})$ -semimodule map, preserves all labels, and preserves the distinction between $\tau = (0, 0_L)$ and $e = (0, 1_L)$. Its image consists of the full carrier on $\text{im } T$; its amplitude-zero inverse image is $G_L(\ker T)$, whereas its inverse image of the

single element τ is only $\{\tau\}$. In particular HCB22 lifts to the exact amplitude-zero inverse-image statement

$$G_L(\partial)^{-1}(\{(0, \lambda) : \lambda \in L\}) = G_L(\mathcal{B}), \quad G_L(\mathcal{E} \oplus \mathcal{B}) \xrightarrow[\cong]{G_L(A)} G_L(\mathcal{D}). \quad (\text{HCB31})$$

The inverse of the last arrow is $G_L(A^{-1})$ and leaves the same label intact. Here $G_L(\mathcal{E} \oplus \mathcal{B})$ is the full carrier with a common label on the ordered pair; it is not silently identified with the product of two independently labelled carriers. In particular, the coefficient cancellation in HCB27 never changes a supported zero into global absence. For an explicit mixed example, write $u = \Psi_{a-1,b} + \Psi_{-a,b}$ with $0 < a < 1$. This is a nonzero function: for $y > b\sqrt{3/(2\pi)}/\min(a, 1-a)$ every term of the two defining series is positive, since $g(x) > 0$ when $x > \sqrt{3/(2\pi)}$. Consequently its only allowed nonzero-amplitude common label is 1_L , and the exact lifted boundary value is

$$G_L(\partial)(u, 1_L) = (0, 1_L) = e \neq \tau = (0, 0_L). \quad (\text{HCB32})$$

This statement concerns membership in the strong test space and the retained common support. It does not identify $\mathcal{B}$ with the restriction image $\mathcal{V}$, or identify a general strong-space function with a cohomologically trivial function. The special actual-image conclusions have their separate explicit source maps in HCB4, HCB25 and HCB26.

The exact retained original-zeta receiving formulas, with parameter recovery, are

$$\begin{aligned} \int_0^\infty R_{t,b}(y) y^{s-1} dy &= b^s \pi^{-s/2} [\zeta(s) - \zeta(s, 1+t)] \left[\Gamma(s/2+2) - \frac{3}{2} \Gamma(s/2+1) \right], \\ \int_0^\infty R_{t,b}(y) \frac{dy}{y} &= -\frac{t}{2}, \quad \int_0^\infty R_{t,b}(y) dy = 0, \quad t = -2 \int_0^\infty R_{t,b}(y) \frac{dy}{y}. \end{aligned} \quad (\text{HCB33})$$

The first equality is an equality of convergent integrals on $\text{Re } s > -2$ with removable values evaluated there, and continues meromorphically with residues $-c_m(t, b)$ at $-2m$. It follows by subtracting the two HCB15 formulas first on $\text{Re } s > 1$, then using HCB18. The two finite-moment equalities subtract HCB20 at $a = 1+t$ from its value at $a = 1$. The last equality gives the original parameter exactly, for every $t > -1$ and $b > 0$. In particular the vanishing of the degree-one difference does not erase its degree-zero response or any boundary coefficient. This receiver acts on the retained pair or on its actual defect coordinate; it cannot be evaluated from the constant sum f_b alone.

6 Rational mixed boundaries with the compact unit coordinate retained

The original restriction map is that of Connes–Consani–Marcolli, *The Weil proof and the geometry of the adeles class space*, source labels `TrMb0`, `rangeVdef`, and `charCK1sum`. The finite character orthogonality and original Hurwitz factors already proved in HCR33–40 are used here; this calculation constructs their actual adelic antecedents and keeps the compact unit coordinate. It does not declare a slice in that coordinate to be an equivariant replacement.

Keep η of ETC1, $b > 0$, an integer $q \geq 3$, and $1 \leq r < q$ with $\gcd(r, q) = 1$. Define the Bruhat test

$$\xi_{r,q,b}(x) = 2\eta(x_\infty/b)\mathbf{1}_{r/q+\widehat{\mathbb{Z}}}(x_f), \quad F_{r,q,b}(u, y) = E\xi_{r,q,b}(u, y). \quad (\text{RMB1})$$

Its original additive value and integral vanish: $\eta(0) = 0$, $\int_{\mathbb{R}} \eta = 0$, and the translated finite lattice has volume one. Hence $F_{r,q,b}$ belongs to the actual restriction image V in SWQ6, before any evaluation or averaging of $u \in K = \widehat{\mathbb{Z}}^\times$.

The contributing rational numbers α in its restriction sum satisfy $\alpha u - r/q \in \widehat{\mathbb{Z}}$. Multiplying by q shows $q\alpha \in \mathbb{Z}$; writing it as n , the remaining condition is $nu \equiv r \pmod{q}$. Therefore

$$F_{r,q,b}(u, y) = 2 \sum_{\substack{n \in \mathbb{Z} \\ n \equiv ru^{-1} \pmod{q}}} \eta\left(\frac{ny}{qb}\right). \quad (\text{RMB2})$$

Here u^{-1} means its actual invertible residue modulo q . The residue is nonzero, so $n = 0$ is absent. Absolute Gaussian convergence justifies every finite regrouping. At the identity, positive and negative integers give exactly

$$F_{r,q,b}(1, y) = \Psi_{r/q-1,b}(y) + \Psi_{-r/q,b}(y). \quad (\text{RMB3})$$

This explains the reflected boundary cancellation of HCB using an original adelic function. It does not yet remove its u coordinate.

For a Dirichlet character modulo q , regarded as a character of K and extended by zero on nonunit integers, define the literal character projector

$$(P_\chi F)(u, y) = \chi(u) \int_K \overline{\chi(v)} F(v, y) dv, \quad \int_K dv = 1. \quad (\text{RMB4})$$

For these level- q functions the integral is the finite average over $(\mathbb{Z}/q\mathbb{Z})^\times$. It preserves the original restriction image: the unit translations of $E\xi$ are E of the correspondingly translated adelic test, and the finite linear average keeps both zero endpoint conditions.

For a fixed unit integer n , the condition $nv = r$ has one solution modulo q of Haar mass $1/\varphi(q)$. Its character value is $\chi(r)\chi(n)$. Thus

$$\begin{aligned} (P_\chi F_{r,q,b})(u, y) &= \frac{2}{\varphi(q)} \overline{\chi(r)} \chi(u) \sum_{n \in \mathbb{Z} \setminus \{0\}} \chi(n) \eta(ny/(qb)) \\ &= \begin{cases} \frac{4}{\varphi(q)} \overline{\chi(r)} \chi(u) \sum_{n \geq 1} \chi(n) \eta(ny/(qb)), & \chi(-1) = 1, \\ 0, & \chi(-1) = -1. \end{cases} \end{aligned} \quad (\text{RMB5})$$

The second line uses the evenness of η and retains the two signs of n . Character orthogonality HCR33 gives the exact inverse $F_{r,q,b} = \sum_{\chi \bmod q} P_\chi F_{r,q,b}$. All sums here are finite in the character index. An explicit projected antecedent, in addition to the finite-average construction, is

$$\xi_\chi(x) = \frac{2}{\varphi(q)} \overline{\chi(r)} \eta(x_\infty/b) \chi(qx_f) \mathbf{1}_{q^{-1}\widehat{\mathbb{Z}}}(x_f), \quad P_\chi F_{r,q,b} = E\xi_\chi. \quad (\text{RMB5a})$$

On the indicated finite support, $\chi(qx_f)$ is the character of its residue in $\mathbb{Z}/q\mathbb{Z}$, extended by zero on nonunits. To check the equation, contributing rationals are $\alpha = n/q$, and $\chi(nu) = \chi(n)\chi(u)$; substitution gives the first line of RMB5 with its exact factor. Both endpoint conditions hold because $\eta(0) = \int \eta = 0$.

Let

$$I_\eta(s) = \frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2). \quad (\text{RMB6})$$

It is the original two-term Mellin integral computed in ETC11, not an unspecified completion. For $\Re s > 1$, absolute Fubini in RMB5 gives, for each even character,

$$\int_0^\infty (P_\chi F_{r,q,b})(u, y) y^s \frac{dy}{y} = \frac{4}{\varphi(q)} \overline{\chi(r)} \chi(u) (qb)^s I_\eta(s) L(s, \chi). \quad (\text{RMB7})$$

All factors, including q^s, b^s , the Haar denominator and the two Gamma terms, remain explicit. Summing at $u = 1$ recovers HCR36 for the two reflected Hurwitz functions. Each $P_\chi F$ lies in $V \subset B$, so its Mellin integral is entire; RMB7 specifies its meromorphic continuation with all exceptional local factors retained.

For the principal character, inclusion and exclusion over the finite set of prime divisors of q gives the stronger original-function identity

$$P_{\chi_0} F_{r,q,b} = \frac{2}{\varphi(q)} \sum_{d|\text{rad}(q)} \mu(d) f_{qb/d}. \quad (\text{RMB8})$$

Indeed $\mathbf{1}_{\gcd(n,q)=1} = \sum_{d|\text{rad}(q), d|n} \mu(d)$. Insert this finite identity into RMB5, put $n = dm$ and use $f_c = 2 \sum_{m \geq 1} \eta(my/c)$. This proves RMB8 without a rearrangement of a divergent endpoint integral. At the Mellin level it retains all bad-prime factors as $L(s, \chi_0) = \zeta(s) \prod_{p|q} (1 - p^{-s})$. For $q = 3, 4, 6$, the only even character is principal, giving the three actual V identities of HCB for the reflected pairs. For other moduli the nonprincipal even summands in RMB5 must also be retained.

The exact replacement for forgetting the unit coordinate is the finite tuple of character components, with inverse just proved. It also preserves convolution: for $F(u, y) = \sum_\chi \chi(u) F_\chi(y)$ and $G(u, y) = \sum_\chi \chi(u) G_\chi(y)$ at the same finite level, compact character orthogonality in the original convolution integral gives

$$(F * G)(u, y) = \sum_\chi \chi(u) (F_\chi *_{\mathbb{R}_+} G_\chi)(y). \quad (\text{RMB9})$$

The real integrals are absolutely convergent by SWQ4. In contrast the slice map $R(F)(y) = F(1, y) = \sum_\chi F_\chi(y)$ introduces all the mixed character products on its target. For a concrete failure take a nontrivial character χ , $F(u, y) = \chi(u) e^{-(\log y)^2}$ and $G(u, y) = e^{-(\log y)^2}$. Then $F * G = 0$ by orthogonality, whereas $(RF * RG)(1) = \int_{\mathbb{R}} e^{-2x^2} dx > 0$. Both inputs belong to the original strong test space. This proves the precise failure of the slice as a convolution morphism, and RMB9 supplies its full invertible replacement.

The failure also occurs on actual elements of V . Take $q = 5, r = 1$ and the even nonprincipal real character χ with $\chi(1) = \chi(4) = 1$ and $\chi(2) = \chi(3) = -1$. Write $P_\chi F(u, y) = \chi(u) A_\chi(y)$ and $P_{\chi_0} F(u, y) = A_0(y)$, and put $g = A_\chi *_{\mathbb{R}_+} A_0$. The two functions

$$X(u, y) = \chi(u) g(y), \quad Y(u, y) = g(y) \quad (\text{RMB10})$$

both belong to V : X is the convolution of $\chi A_\chi \in V$ with $\chi A_0 \in B$, while Y is the convolution of $A_0 \in V$ with the radial $A_\chi \in B$. Use SWQ14's proved ideal property. Their full convolution is zero by character orthogonality. However RMB6 gives $I_\eta(2) = 1/(4\pi)$ and RMB7 gives $MA_\chi(2) = 25b^2 L(2, \chi)/(4\pi)$ and $MA_0(2) = 25b^2 L(2, \chi_0)/(4\pi)$. Both L values are positive: their absolutely convergent Euler products have positive factors $(1 - \chi(p)p^{-2})^{-1}$, and $\sum_p p^{-2} < \infty$ keeps their product nonzero. Thus $Mg(2) > 0$ and $M(g *_{\mathbb{R}_+} g)(2) = Mg(2)^2 > 0$. The identity

slice has therefore changed a zero product of two actual trivial correspondences into a nonzero product. The retained character tuple in RMB9 preserves their zero product exactly.

On the original full support carrier, apply the common-label lift of the character decomposition and of its inverse to the entire finite direct sum of component function spaces. Both maps are complex-linear; they preserve label joins, the scalar action and inverse compositions. By RMB9 they also preserve the specified convolution products, with label meet. Thus the corrected finite-character receiver retains every label, including e and τ ; it does not assign independent labels to components of one common-label input. This exact map neither asserts that an identity slice lies in the radial restriction ideal nor changes which character L -function appears in RMB7.